

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifica-
8 tion demonstrating how cognitive architectures can emerge as attrac-
9 tor states from established physical primitives, without introducing
10 new fundamental physics. Starting from the finiteness of a localized
11 system’s phase space, we construct a framework for predictive process-
12 ing; the spacetime symmetries are the setting rather than a premise,
13 and no theorem consumes them. Spontaneous symmetry breaking gener-
14 ates necessary topological boundaries, and finite phase space sub-
15 jects these boundaries to Landauer’s thermodynamic erasure. Agency

16 arises when systems evolve along paths of least action, minimizing en-
 17 tropy production via internal structural resonance with external per-
 18 turbations. Scaling this dynamic to macroscopic levels aligns with
 19 emerging paradigms in neuroscience—specifically classical electrody-
 20 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 21 tion, culminating in a topological fixed point: a unified “Self.” We
 22 state the claim narrowly, because the subject matter invites a wider
 23 reading than the work supports: no new physics is derived here. Every
 24 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 25 ble’s classification of topological defects, Landauer’s erasure bound,
 26 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-
 27 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed
 28 point. What is new is that they are stated in one formal language,
 29 their consistency mechanically checked, and the points where the chain
 30 relies on stipulation rather than derivation located exactly. The chain
 31 itself composes to a single theorem, whose eight unproved steps are
 32 its arguments rather than sentences about it, and which are jointly
 33 satisfiable rather than jointly empty. The development declares no ax-
 34 ioms and every structure carrying physical content is inhabited by a
 35 witness; the synchronization step is proved from a dynamics for initial
 36 data inside an explicit basin and assumed outside it, and the identi-
 37 fication of a global section of the probability sheaf with the unity of
 38 experience is a stipulation, marked as such throughout rather than
 39 absorbed into the derivation. The step from a coarse-grained cou-

pling to the brain’s endogenous electromagnetic field is likewise an empirical commitment and not a deduction; we separate it from the coarse-graining theorem that precedes it, weaken it to a modulatory-but-integrative role, and give four conditions that would falsify it. The framework is nonetheless falsifiable in one place that is its own and no rival’s: recovery from sudden awakening should follow a delayed sigmoid rather than an exponential, because the variable gating it is astrocytic volume on a timescale of seconds to minutes rather than cortical phase realignment on a timescale of milliseconds; and, more sharply, the concentration and the coherence of the phase distribution must collapse onto a single parameter-free curve, the Bessel ratio, throughout that recovery. As a corollary, we show why rigid von Neumann architectures (e.g., GPUs) are thermodynamically disadvantaged or structurally constrained from reaching these specific continuous thermodynamic attractors.

1 Introduction

The pursuit of a physical theory of consciousness typically involves searching for the biological correlates of mind or treating subjective experience as an abstract computational algorithm. This paper asks a more fundamental question: If we assume only basic physical primitives—continuous symmetries (the Poincaré group), geometry, and thermodynamics—is the emergence of a unified, perceiving entity a natural physical attractor?

62 We propose a framework where the structural properties of consciousness
 63 are derived as physical attractor states. Establishing a deductive chain from
 64 basic principles, we demonstrate how continuous symmetries break to form
 65 boundaries, how those boundaries dissipate free energy as non-equilibrium
 66 steady states, and how their evolution along paths of least action results
 67 in physical resonance with their environment. This eventually scales into
 68 the macroscopic electromagnetic unity proposed by emerging field theories
 69 of mind. To avoid critiques of “math theatre,” we delineate between literal
 70 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 71 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 72 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 73 spective aligned with Russellian Monism: objective measurements of macro-
 74 scopic electromagnetic fields constitute the third-person “outside,” while the
 75 system’s internal physical deformation to reach structural resonance consti-
 76 tutes the first-person intrinsic experience.

77 **What is new here, and what is not**

78 A framework paper about consciousness invites a particular suspicion, and it
 79 is better answered at the start than at the end. **No new physics is derived**
 80 **here.** Every physical result the chain invokes is established, and most of
 81 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 82 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 83 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound

84 on the heat cost of erasure (Landauer, 1961); Gibbs' inequality for the relative
 85 entropy (Cover & Thomas, 2006); Barbalat's lemma for the asymptotics of a
 86 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
 87 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 88 Bell and Crooks relating dissipation to nonpredictive information (Still et
 89 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 90 dissipated work with the relative entropy between forward and time-reversed
 91 ensembles (Kawai et al., 2007); and Banach's fixed-point theorem (Banach,
 92 1922). A physicist reading this paper will recognize all of it and would
 93 establish the point in three paragraphs if we did not. Conceding it costs
 94 nothing.

95 What is new is of a different kind, and it is a smaller claim than a theory
 96 of consciousness usually makes. These results have been stated in one formal
 97 language, their mutual consistency has been mechanically checked, and the
 98 places where the chain relies on stipulation rather than derivation have been
 99 located exactly. The value of a machine-checked chain is not that any link
 100 is novel; it is that the claim being made *about* the chain is itself checkable,
 101 and we state it in the form a reader can check rather than in the form that
 102 asks to be believed. Four things. The development declares **no axioms**:
 103 **#print axioms** on any result reports only **propext**, **Classical.choice** and
 104 **Quot.sound**. Every physical postulate is a field of a class that each model
 105 discharges for itself rather than a global assertion, and every such class is
 106 inhabited by an explicit witness, so no theorem is vacuous for want of a

107 model. The links form a single *conditional composition theorem* — **chain**,
 108 in **Chain.lean** — with **eight named hypotheses**: four formalization gaps,
 109 two modelling assumptions and two physical commitments. Its conclusion is
 110 **UnifiedSelf**: the particular section produced by sheaf gluing is the unique
 111 fixed point of self-prediction, rather than Unity and Self being witnessed by
 112 unrelated sections. The $n8 \rightarrow n9$ hypothesis explicitly carries both the non-
 113 linear Lipschitz law and this state identification; neither is derived from field
 114 dynamics. The eight hypotheses are jointly satisfiable on one toy substrate
 115 (**chain_nonvacuous**), which rules out inconsistency but is not evidence that
 116 one cortical mechanism realizes the composition. The seams — where physics
 117 stops and interpretation begins — are thereby fixed in the source rather than
 118 negotiable in the prose.

119 Composing the chain also refuted two of its own arrows, and Figure 1
 120 draws the corrected picture rather than repairing it quietly. The capacity
 121 bound of the first link has no hypotheses at all, so an implication from it to
 122 the existence of a boundary is equivalent to assuming the boundary outright:
 123 those two links are independent starting points and not consecutive steps.
 124 What finiteness does imply directly is the heat cost, skipping the boundary
 125 entirely. Neither of the two arrows that are theorems is one the chain’s
 126 usual narration draws. Section 2.1 reports what the check cost: three of five
 127 original axioms each proved **False**, from one shared root cause, and none of
 128 the three had been caught by reading.

129 The framework is not thereby unfalsifiable, and one of its predictions is

130 its own rather than a rival's. Section 11 argues that recovery from sudden
 131 awakening should follow a *delayed sigmoid* rather than an exponential, be-
 132 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
 133 clearance, on a timescale of seconds to minutes, while the cortical rhythms
 134 whose realignment is usually taken to constitute the recovery move in mil-
 135 liseconds. That is a claim about functional form; it is fittable against data
 136 of a kind already collected; and a purely electrical account of the same re-
 137 covery predicts a different form. Because a delayed sigmoid is what any
 138 slow saturating gate would give, the section also states the sharper claim the
 139 same mathematics supports: at every instant of the recovery the concentra-
 140 tion and the coherence of the phase distribution must lie on the fixed curve
 141 $r = I_1(a)/I_0(a)$, the same curve for every subject, with no free parameters
 142 and with the threshold read off the ratio a/r rather than fitted.

143 **2 Formalization in Lean 4**

144 A core contribution of this work is the verification of our deductive chain
 145 using the Lean 4 interactive theorem prover. This approach builds on recent
 146 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
 147 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
 148 2026). It is necessary to explicitly distinguish which claims have been math-
 149 ematically derived as theorems within the proof assistant, and which remain
 150 irreducible physical postulates. Figure 1 shows the chain with the status of

every node *and of every arrow* marked, and Table 1 states in seven rows what each link assumes. The complete claim-by-claim map to Lean identifiers, which runs to several pages, is Table S1 in the supplement; the main text does not depend on reading it.

2.1 Soundness of the formalization

Formalization is only as trustworthy as the axioms it rests on, and an axiom that is inconsistent silently makes every theorem downstream of it vacuous while leaving the build green. We report a case of exactly this, found by auditing our own development, because the failure mode is easy to reproduce and we have not seen it discussed in the applied-formalization literature.

The failure has one shape, and naming it is more useful than cataloguing its instances: *an axiom that constrains a symbol it does not itself bind*. Such a declaration is not a postulate about one system. It is a constraint on every model of the class the symbol belongs to, including models built to violate it, and a class that has such a model is thereby inconsistent. Three concrete cases, each of which proves `False` on its own:

- `spontaneous_symmetry_breaking_pointwise_min` took the potential-energy functional as an *implicit* argument occurring only in its hypothesis `PotentialEnergy phi = V v0`. Instantiating that functional with the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$

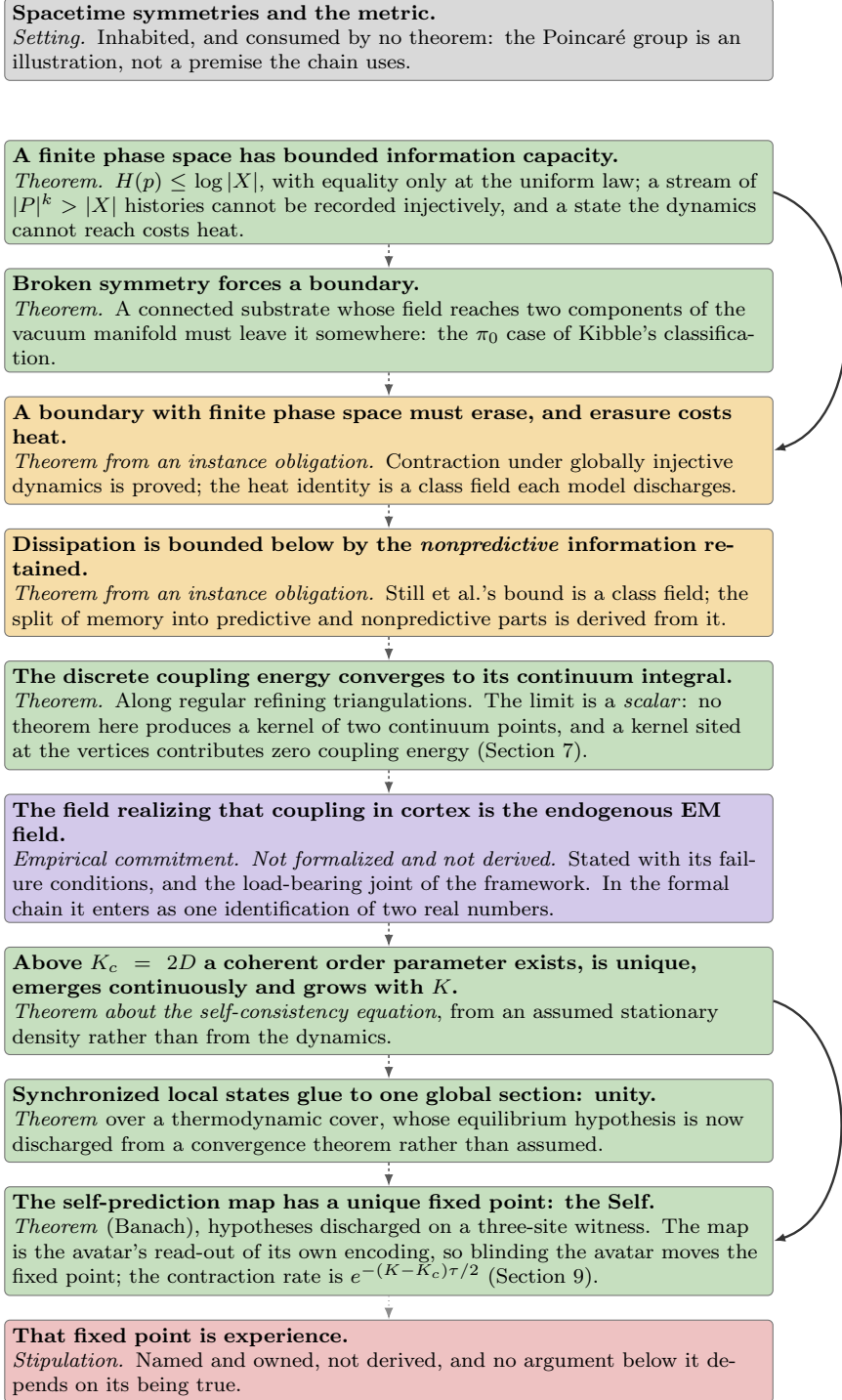


Figure 1: The deductive chain, with⁹the status of each node and of each arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of chain (Chain.lean): an implication between the two node statements which the development does not prove and which is therefore an argument the theorem

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasures costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat.eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, and synchronized patches glue into a unique global section	Theorem 10	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum whose synchronized patches agree on overlaps; the first is derived from a convergent trajectory, the second is an instance obligation throughout.
The coherent order parameter carries a unique reflexive fixed point:	Theorem	The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate

172 this proves $1 = 0$.

- 173 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
174 a bath-entropy difference *for every instance* of the class. An instance
175 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- 176 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P\|Q)/\Delta t$ for *all* P and Q , while σ
177 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
178 point mass against a uniform distribution refutes it.

179 Two further declarations of the same shape are worth recording be-
180 cause their refutation is different in kind. `phase_invariant_periodic` and
181 `sync_to_section_eq`, as global assertions pinning free fields of `LocalSectionSynchronization`,
182 are refutable as soon as the probability presheaf admits two distinct global
183 sections — so the refutation needs no arithmetic at all, only that the sheaf is
184 not trivial. Neither is a postulate in the development: the periodicity condi-
185 tion is an obligation of the constructor that builds a cover from an invariant
186 measure, and `sync_to_section_eq` is a theorem about the global section the
187 sheaf condition produces (Derivation 5).

188 The rule that prevents it is uniform, and we state it as the rule we follow:
189 *a physical postulate that mentions a class field must be a field of that class,*
190 *never a standalone axiom quantified over all instances.* A postulate is then
191 an obligation each model discharges, not a global claim about every model.
192 Under that rule the development declares no axioms at all. Two of the
193 statements above are not postulates in the first place — pointwise vacuum

194 minimization is proved from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs'
 195 inequality is Mathlib's — and the genuinely physical ones, Landauer's heat
 196 identity and the two information bounds, are instance obligations.

197 This makes non-vacuity the live question, since an uninhabitable class
 198 makes its theorems vacuous just as surely as an inconsistent axiom did.
 199 `Examples.lean` therefore constructs witnesses: a one-bit erasure system with
 200 a genuine bath that discharges `heat_eq` and reproduces $\Delta Q = k_B T \ln 2$; a
 201 perfectly resonant system discharging `kl_bound`; and a scalar field on a one-
 202 point spacetime discharging `ActionPrinciples`. Derivation 5's gluing struc-
 203 tures are the most demanding case and are witnessed too: `Examples.lean`
 204 builds a `ThermodynamicCover` over a three-site substrate carrying the dis-
 205 crete topology and its Borel σ -algebra, covered by two *overlapping* open
 206 patches whose intersection is a single shared site. The local sections are
 207 germ families of a phase-indexed finite measure $\mu_\theta = (1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed
 208 into the sheafification of the probability presheaf; the family is provably
 209 non-constant in θ (`phaseMeasure_not_const`), so the periodicity obligation
 210 of the constructor `LocalSectionSynchronization.ofInvariantMeasure` is
 211 discharged by an argument about `cos` rather than satisfied trivially, and
 212 `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 213 rather than assumed. `global_section_from_thermodynamics` then fires on
 214 this instance and glues the two patch-local sections into a unique global one.
 215 Two further covers answer the follow-up question of how non-degenerate such
 216 a witness can be made. `Examples.lean` §13 has a negative half worth record-

217 ing: the phase field can be made non-constant as a function, but not as a
 218 physical state, because `thermodynamic_equilibrium` forces every instance
 219 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 220 round, from two patches with independently chosen mass profiles that meet
 221 only on the site they share, and the section the gluing returns is neither of
 222 them. That is the same kind of result as `contracting_implies_const` in
 223 Derivation 6 — a proof that the witness is as strong as the class permits,
 224 rather than a witness that could be improved by trying harder.

225 What the witnesses establish is that the classes are inhabitable, so the
 226 theorems about them are not vacuous. What they do not establish is that
 227 a cortex is such an instance. In the cover above the phase field is already
 228 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 229 the equilibrium obligation dischargeable; getting a physical system into that
 230 configuration is the content of the informal dynamical argument, not of the
 231 Lean development.

232 The same audit turned up a defect of a different kind, worth record-
 233 ing because it is not an axiom problem and would not have been caught
 234 by `#print axioms.mesh_refinement_convergence` — the claim that dis-
 235 crete edge weights converge to the continuous energy as the mesh is refined
 236 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
 237 not merely unproved but false. Nothing required the edge regions to cover
 238 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
 239 all empty satisfied the mesh hypothesis while contributing nothing to the

240 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
 241 quantified over a triangulation and then resolved the vertex type by instance
 242 search rather than through the bound variable, so the variable was never
 243 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
 244 gence is now stated along a *sequence* of triangulations of a named region
 245 S , each required to be regular — edge regions measurable, without self-
 246 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 247 plicial data in the sense that a vertex lies in the closure of every region
 248 of an edge at it. Closure rather than membership is forced: cells are dis-
 249 joint, so a shared vertex belongs to at most one of them. The manuscript’s
 250 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 251 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 252 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 253 tion, reading the double sum as a Riemann sum would be a category error.
 254 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 255 the integrand, and convergence follows for uniformly continuous integrands
 256 as the mesh shrinks. The regularity conditions pull against each other, so
 257 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 258 half-open cells and checks that the approximation genuinely moves — $\frac{1}{2}$ at
 259 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 260 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 261 the accompanying simulation measures; that would need a midpoint error
 262 term with a second derivative, which we have not developed.

263 A third defect was one of bookkeeping across files, and it is the kind that
 264 survives any number of green builds because each half is individually cor-
 265 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 266 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 267 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 268 — phase-locked in both directions — and everything downstream of it, in-
 269 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 270 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 271 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 272 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 273 it has no minimum for phase-locking to attain, and the sentence “the phase-
 274 locked state minimizes the Lyapunov potential” was, read literally, false of
 275 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 276 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 277 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 278 frequency system — which is exactly the system Derivation 5 already worked
 279 with — the two potentials coincide there, and the descent result transports
 280 onto the coupling potential whose minimizers are the phase-locked states.
 281 Phase differences, phase-locking and the magnitude of the order parameter
 282 are all proved invariant under the change of frame, so nothing observable
 283 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 284 two oscillators with a common frequency, discharging every hypothesis. The
 285 restriction is real and we state it plainly: the reduction is exact only for

286 identical frequencies. A genuine spread of natural frequencies leaves residual
 287 detunings in the reduced system, phase-locking then requires coupling above
 288 a critical K_c , and that K_c — Kuramoto’s threshold for a spread of frequen-
 289 cies, not the noise threshold $2D$ of Section 10 — is nowhere in our Lean
 290 development. It is studied numerically, in Figure 5, as Table S1 records.

291 A fourth defect was dimensional, and it concerned the one row of Table S1
 292 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 293 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 294 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 295 against an unnormalized measure carries the substrate’s mass twice over, so
 296 a constant kernel K on a substrate of total mass m contributed Km^2 . The
 297 comparison therefore depended on how large the substrate was rather than
 298 on how strongly it was coupled, and any kernel whatever could be pushed
 299 above threshold by inflating m . `Examples.lean` §9 exhibits the failure con-
 300 cretely: on a two-site substrate carrying counting measure, a system coupled
 301 at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two *below* threshold
 302 — has an unnormalized double integral of $2 > 1$, and would satisfy any pred-
 303 icate stated with that integral. The fix is to require the substrate’s measure
 304 to be a probability measure, which is the normalization in which the mean-
 305 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 306 then gives that a constant kernel has mean-field strength exactly K , and
 307 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 308 We flag what this does and does not do. It makes the predicate say something

309 whose truth depends only on the physics, and §9 witnesses it on both sides of
 310 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 311 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 312 at $K = 2D$, which is a separate question from whether the predicate is di-
 313 mensionally sound. That bifurcation is proved below, in both directions,
 314 from an assumed stationary density; deriving the density itself needs the
 315 Fokker–Planck operator and is not done.

316 We draw no grand moral from this beyond the obvious one. The value
 317 of a proof assistant here was not that it certified our argument, but that it
 318 made a specific, checkable claim about what had been assumed — and that
 319 claim turned out to be false in a way no amount of reading the prose would
 320 have revealed.

321 **3 Axiom 1: Symmetries as Setting, and the** 322 **Capacity of a Finite Phase Space**

323 Our starting premise relies on fundamental geometry. In our Lean formaliza-
 324 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the
 325 Poincaré group) act upon field states, and the invariance of the system’s total
 326 energy under that action is recorded by the class `SymmetryInvariantAction`.
 327 Physically, it is such an invariance that Noether’s theorem converts into the
 328 conservation of energy and momentum. The scope of the *formal* claim is nar-
 329 rower and we state it plainly: the invariance is a class field, i.e. an assumption

330 a model must discharge, and no conserved quantity is ever constructed from
 331 it. There is no Noether theorem in our Lean, and nothing downstream of this
 332 section depends on one. The class is now inhabited — `Examples.lean` §11
 333 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
 334 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
 335 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
 336 changed by the symmetry, so an energy-minimising field and its image have
 337 equal energy and differ. That is symmetry breaking exhibited rather than
 338 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 339 no one-parameter family to differentiate along in the first place. Simultane-
 340 ously, discrete permutation symmetries dictate the statistical mechanics of
 341 any physical system, defining a strictly finite volume of phase space.¹

342 Because phase space is finite, any localized physical system possesses a
 343 mathematical limit on the amount of information (distinguishable states) it
 344 can embody at any given moment. That sentence is the section’s contribution
 345 to the chain, and three theorems carry it, in `Phase1_PhaseSpaceCapacity.lean`,
 346 whose only hypothesis about the system is that its state space is finite: no

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
 amples, the underlying Lean formalization does not hardcode specific physics. It defines
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-
 ics). What the formalization supplies is the vocabulary in which symmetries, conserved
 quantities and phase-space constraints can be *stated*; it does not derive them. All three
 of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its in-
 variance in §11, the real line with the Euclidean metric in §12 — but the instances are
 witnesses of inhabitability, not of physics: no theorem in the development consumes a
 symmetry group or a metric, and that half of the premise is setting rather than content.
 The other half of the premise is not setting, and the paragraphs below are what it does.

347 symmetry group, no metric, no action functional appears anywhere in them.

348 The first is the capacity bound itself. A probability distribution p on a
349 finite state space X satisfies $H(p) = -\sum_x p(x) \log p(x) \leq \log |X|$ (`shannon_`
350 `entropy_le_log_card`), with equality if and only if p is uniform (`shannon_`
351 `entropy_eq_log_card_iff`). The “only if” half is the part worth stating: it
352 says $\log |X|$ is the exact capacity rather than a convenient over-estimate, so
353 a system holding any distribution other than the flat one over its states holds
354 strictly less than its size suggests. The proof is Gibbs’ inequality against the
355 uniform law — $\log x \leq x - 1$ applied at $x = 1/(|X|p(x))$, with the $p(x) = 0$
356 case carrying the slack that makes the equality condition sharp — and it is
357 elementary. What was missing here was the statement, not the difficulty.

358 The second says the limit binds. Let the boundary absorb a stream of
359 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
360 possible histories. Once $|X| < |P|^k$, the map from histories to final states
361 cannot be injective (`absorb_not_injective`), so two distinct histories leave
362 the system in the same state, and the entropy of the source strictly exceeds
363 anything the system can hold (`source_entropy_exceeds_capacity`). The
364 update map is arbitrary and is allowed to be reversible at every individual
365 step; the collision is forced by counting alone, which is the sense in which the
366 premise, and not the model of the dynamics, is doing the work.

367 The third is what Derivation 2 consumes, and it is the one place where
368 finiteness does something no other hypothesis could. On a finite state space,
369 a map that fails to reach every state is not injective (`is_erasure_of_not_`

370 `surjective`) — which is exactly the erasure condition Landauer’s bound re-
 371 quires — so any update of a finite phase space leaving some state unreachable
 372 dissipates strictly positive heat (`finite_phase_space_dissipates`). On an
 373 infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses 0 and
 374 is injective. `Examples.lean` §19 records that failure next to the witnesses, so
 375 the finiteness hypothesis is fenced rather than assumed to be load-bearing,
 376 and runs all three results on a single bit: the fair coin attains $\log 2$, a $3/4$ -
 377 $1/4$ bit falls strictly short, two binary perturbations already outrun a one-bit
 378 memory, and §1’s one-bit thermodynamic system pays heat for a state it
 379 cannot reach.

380 None of this derives the finiteness of phase space. Finiteness is assumed,
 381 as it is in any statistical-mechanical treatment, and the Poincaré group re-
 382 mains an illustration in prose. What has been removed is the gap between
 383 assuming the premise and using it.

384 **4 Derivation 1: Symmetry Breaking and the** 385 **Inevitability of Boundaries**

386 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 387 exist. For any dynamic process to occur, the continuous symmetries of the
 388 background field must undergo spontaneous symmetry breaking.

389 When a field drops to a lower, asymmetric energy state, topology dic-
 390 tates the natural creation of defects (such as domain walls or monopoles).

391 This constraint necessitates the creation of a physical boundary—a localized
 392 topological distinction between an “inside” and an “outside.” While such
 393 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
 394 entity, it is not sufficient for agency. The boundary is relational, defined by
 395 the twisted geometry of the field surrounding it, and must be coupled with
 396 non-equilibrium thermodynamics to exhibit active behaviors.

397 That first sentence is a claim about topology, and it needs a different
 398 argument from the energy claim beside it. The energy half — an energy-
 399 minimizing field sits in the vacuum manifold almost everywhere (`pointwise_`
 400 `vacuum_of_global_min`) — together with the observation that a boundary
 401 defect *obstructs* extension over a contractible interior, produces no defect:
 402 both are statements about a field already given. What produces one is
 403 `exists_notMem_of_connectedComponentIn`: if the substrate is connected
 404 and a continuous field takes values in two *different connected components*
 405 of the vacuum manifold M , then somewhere the field is not in M at all.
 406 That excluded region is the wall. This is the π_0 case of Kibble’s classifica-
 407 tion of topological defects by the homotopy of the vacuum manifold (Kibble,
 408 1976), and the proof is elementary in exactly the way it is elementary in the
 409 physics literature: the continuous image of a connected space is connected, a
 410 connected subset of M lies inside one component of M , and the field’s range
 411 would be such a subset if it never left M . No hypothesis mentions energy, a
 412 potential, or a symmetry; connectedness of the substrate and disconnected-
 413 ness of the vacuum are the whole of it.

414 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 415 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 416 classification and Mathlib’s homotopy-group library would support at least
 417 the next one; they are available and not taken, because the argument this
 418 paper runs uses domain walls and nothing else. The witness is the double
 419 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be
 420 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 421 minima, and its disconnection is proved in the form the theorem consumes:
 422 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 423 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
 424 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field
 425 on *any* connected substrate that reaches both minima, with no formula for
 426 the field anywhere in the statement. That the class is non-empty is checked
 427 separately, because a theorem quantified over all fields is worthless if none
 428 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 429 forced to have sits at the origin (`kink.zero_notMem`).

430 **5 Derivation 2: The Thermodynamic Con-** 431 **straint**

432 This newly formed topological boundary is subjected to gauge perturbations
 433 from the “outside” environment. Because the internal phase space is finite
 434 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations

435 and must overwrite old internal states.

436 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
437 formation irreversibly decreases internal entropy, mandating that the sys-
438 tem exhaust heat into the environment. Philosophers of physics (such as
439 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
440 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
441 (a many-to-one mapping in abstract computational states) with *thermody-*
442 *namic* erasure (a many-to-one mapping in physical phase space). They argue
443 that logical erasure could conceptually be implemented by an injective phys-
444 ical process without dissipating heat.

445 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
446 addresses this critique by defining erasure as a non-injective mapping on the
447 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
448 tion of the bipartite universe (System + Bath) as strictly injective, repre-
449 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
450 to abstract logical erasure, but is a mandatory mathematical consequence
451 of any local physical state space contraction under globally reversible dy-
452 namics. If a cognitive system’s internal topology forces a contraction of its
453 physical state space to accommodate structural resonance, heat dissipation
454 becomes a thermodynamic requirement. The finiteness premise enters at ex-
455 actly this point. What a modeller can usually check about a physical update
456 is not that it is many-to-one but that some state has become unreachable;
457 on a finite phase space those are the same condition, and `finite_phase_`

458 `space_dissipates` is the resulting statement. That identification preserves
 459 the Norton–Shenker distinction rather than blurring it: an update that keeps
 460 the incoming perturbation on the record is injective and dissipates nothing
 461 (`keepRecord_injective`), while one that refreshes the register holding it is
 462 not (`absorbStep_is_erasure`). Losing *which* history occurred is forced by
 463 counting alone and is free; overwriting the register that held it is what is
 464 charged $k_B T \ln 2$. A stable boundary thus constitutes a dissipative structure
 465 (Prigogine & Stengers, 1984), operating as a non-equilibrium steady state
 466 that consumes free energy to pay the thermodynamic cost of erasure. Contin-
 467 uous thermal fluctuations play a constructive role in establishing boundaries
 468 between states while contributing to the system’s flexible, adaptive response.

469 **6 Derivation 3: Prediction as the Thermody-** 470 **namic Cost of Memory**

471 A dissipative structure is driven by its environment. Write X_t for the state
 472 of the structure and S_t for the environmental signal driving it, and let the
 473 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
 474 does not act. Two quantities are then defined at each step: the *memory*
 475 $I(X_t; S_t)$, the mutual information between the structure’s present state and
 476 the signal’s present value, and the *predictive information* $I(X_t; S_{t+1})$, which
 477 is what that same state says about the signal’s next value.

478 **What the entropy-production bound does not give.** There is a nat-
 479 ural route from entropy production to predictive processing, and it does
 480 not work. Model the environment as a stochastic driving force $P(\text{ext})$ and
 481 the structure's internal transitions as $Q(\text{int})$. Entropy production is then
 482 bounded below by $D_{KL}(P||Q)/\Delta t$ — this is Gibbs' inequality, it is carried
 483 in Lean as an instance obligation, and §3 of the supplement states it. Re-
 484 arranged it reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so a system descending on σ lowers
 485 a *bound* on the divergence, and it is tempting to conclude that the internal
 486 transition probabilities come to mirror the environment's statistical symme-
 487 tries. They do not, because a driven system does not relax to $\sigma = 0$. It
 488 relaxes to a non-equilibrium steady state at strictly positive entropy produc-
 489 tion, and the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number.
 490 The inequality is true and does real work — combined with Gibbs it gives
 491 $\sigma \geq 0$ — but no limit may be read off it, and a framework that wants predic-
 492 tive processing has to get it from somewhere else. It gets it from the bound
 493 below.

494 **Where predictive processing comes from: a published bound.** Still,
 495 Sivak, Bell and Crooks (Still et al., 2012) prove that for a system driven by
 496 an environmental signal the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

497 The bracket is the *nonpredictive* information: what the structure retains
 498 about the signal's present that says nothing about the signal's future. Dis-
 499 sipation is bounded below by it. Memory that fails to predict is not merely
 500 useless to the structure; it is thermodynamic waste, and the waste is quan-
 501 tified in energy.

502 **Three consequences, and they are theorems rather than readings.**

503 The bound is carried in Lean as a class field, `PredictiveDissipation.`
 504 `still_bound (Phase3_PredictiveThermodynamics)`, for the same reason ev-
 505 ery other physical postulate here is: its `axiom` form, quantified over all joint
 506 laws with the dissipated work fixed, is refutable, and **Examples §18.4** refutes
 507 it. What is *derived* is everything downstream.

508 First, *the bracket is non-negative*, and this is proved rather than assumed.
 509 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
 510 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
 511 $I(X_t; S_t)$: a system cannot know more about the signal's future than about its
 512 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback-
 513 Leibler divergence but no mutual information, so the latter is defined here
 514 as the divergence of the joint law from the product of its marginals, which is
 515 the standard definition and makes the library's data processing lemma apply
 516 directly.

517 Second, *the second law follows* instead of being assumed. Dissipation is
 518 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$

519 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays
 520 for the entropy-production bound, with one difference that matters: Gibbs'
 521 inequality compares a system to an arbitrary reference distribution, whereas
 522 the data processing inequality compares the system to its own future, which
 523 is a quantity selection can act on.

524 Third, and this is the point of the replacement, *the inference now runs*
 525 *in the valid direction*. The quantity of interest sits on the small side of
 526 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
 527 Any bound on dissipation is immediately a bound on wasted memory, so a
 528 structure held to an energy budget is thereby held to a prediction standard:
 529 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case
 530 $W_{\text{diss}} = 0$, the theorem `predictive_eq_of_no_dissipation` gives that every
 531 retained bit is predictive. This is a weaker statement than the one it replaces,
 532 and it is the one that is true.

533 **The hypothesis is load-bearing, and we exhibit what breaks with-**
 534 **out it.** The data processing step requires that the structure not write to
 535 its environment. In the formalization this is not an assumption stated beside
 536 the theorem but the shape of the evolution itself: the signal's step is applied
 537 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the
 538 system coordinate and through κ on the signal coordinate, so no channel
 539 from X_t to S_{t+1} exists. **Examples** §18.3 shows the restriction is not decora-
 540 tion: a Markov kernel on the *pair* which copies the system's state into the

541 signal carries an independent two-bit law to a perfectly correlated one, rais-
542 ing mutual information from zero to a positive value. An agent that steers
543 its environment can indeed know more about that environment’s future than
544 about its present, and Still’s bound is not about such an agent.

545 **Both regimes are exhibited, not just described.** Examples §18 builds
546 two systems over one correlated two-bit joint law, differing only in the signal’s
547 dynamics. When the signal is frozen, every remembered bit is a predictive
548 bit, the nonpredictive information is zero, and the instance is permitted to
549 dissipate nothing — with the memory itself proved strictly positive, so the
550 conclusion is about a system that does remember something. When the signal
551 is redrawn from its stationary law at each step, the predictive information is
552 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
553 to dissipate $k_B T$ times its whole mutual information. Establishing that the
554 mutual information of the witness is neither zero nor infinite is where the
555 work of that section lies.

556 **An equality, where the entropy-production route had only a bound.**
557 Kawai, Parrondo and Van den Broeck (Kawai et al., 2007) show that the
558 average dissipated work is exactly $k_B T$ times the relative entropy between the
559 forward and time-reversed phase-space densities. Carried here as a predicate
560 on an instance rather than as a further postulate, it replaces the abstract
561 dissipation field with a quantity an experiment can estimate — the statistical
562 distinguishability of a recording of the system from the same recording run

563 backwards — and yields the statement that the irreversibility of the dynamics
564 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
565 `le_arrow`).

566 **What this does not establish.** It does not deliver the Free Energy Prin-
567 ciple. Still’s bound says that prediction is thermodynamically favoured; it
568 says nothing about hierarchical generative models, variational free energy,
569 or Bayesian inference, and we do not claim to have derived any of them.
570 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
571 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-
572 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
573 proved here, which is a weaker relation than derivation and the only one
574 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
575 Maximal predictive information is compatible with the structure storing a
576 lossy, compressed or otherwise unrecognisable function of the signal; a state
577 that predicts is not thereby a state that means. The narrower claim is the
578 one worth making: whatever internal states a physical structure holds, those
579 that do not help predict its environment are paid for in heat, so the pressure
580 toward states indexed to the environment is energetic rather than semantic.
581 We continue to call the phenomenon *structural resonance*, and there is no
582 teleological drive to “survive” in it: predicting the environment is simply the
583 cheaper of the two ways to hold a memory.

584 7 Macroscopic Scaling: A Theorem and a 585 Commitment

586 This is the chain’s fourth step, and it is deliberately not numbered as a
587 derivation. Two steps of different kinds meet here and are easy to run to-
588 gether: a coarse-graining argument, which is a piece of mathematics, and
589 the identification of the coarse-grained medium with the brain’s endogenous
590 electromagnetic field, which is an empirical claim about cortex. Presenting
591 the second as though it inherited the licence of the first would make it look
592 like a deduction, and it is not one. Kept apart, the empirical claim is the
593 framework’s central commitment and the place where it is most easily wrong
594 — which is where a claim of this kind belongs.

595 The formal half: coarse-graining is a theorem, and a 596 narrow one

597 A network of discretely coupled units has a coupling energy that is a sym-
598 metric double sum over pairs. `Phase2.MeshConvergence.lean` proves that
599 this sum converges to a continuous integral: along a sequence of regular
600 triangulations of a region S whose edge regions shrink to points, the dis-
601 crete energy $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$ converges to $\int_S f d\mu$ (`mesh_refinement_`
602 `convergence`), witnessed on the uniform partition of $[0, 1)$ in `Examples §6`.
603 The identification that makes this a coarse-graining rather than an arbitrary

604 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,
605 the double sum is exactly the midpoint rule over unordered edges, so the
606 discrete energy is a quadrature of the continuous one rather than a different
607 quantity that happens to converge to it.

608 What this establishes is worth stating exactly, because it is less than the
609 phrase “coarse-graining, governed by the Renormalization Group” suggests.
610 It is a statement about an *energy functional*, not about a dynamics: nothing
611 here shows that trajectories of the discrete Kuramoto system converge
612 to solutions of a field equation, and the earlier appeal to the Renormaliza-
613 tion Group was decorative — no renormalization group flow is constructed
614 anywhere in this development. The limit object is a scalar density on the sub-
615 strate, in the intended reading the magnitude of the stress-energy tensor, and
616 it is not an electromagnetic field. The continuous object the later sections
617 use, `ContinuousNeuralField`, is *posited* as a structure with a coupling kernel
618 K ; no theorem produces it from the discrete system. The gap between the
619 finite Kuramoto model and its continuum limit is the propagation-of-chaos
620 problem, which we have not addressed and record as open.

621 **The empirical half: which physical field realizes the ker-** 622 **nel**

623 The formal half says that a large network of symmetrically coupled units
624 admits a continuum description with a symmetric kernel. It says nothing

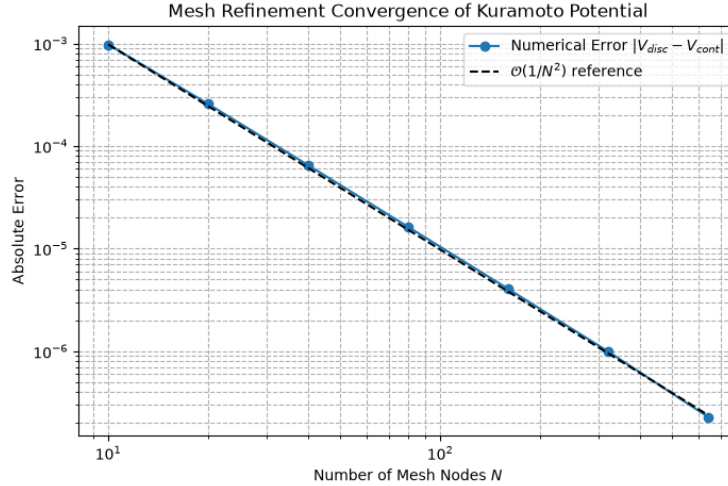


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones.

625 about which physical field in cortex that kernel describes. Our answer is an
 626 empirical commitment, and we state it as one:

627 *In the mammalian cortex, the field realizing the coarse-grained*
 628 *coupling kernel is the endogenous electromagnetic field.*

629 This is a claim about biology, not a consequence of anything above it.
 630 It is what makes the framework falsifiable and it is what we expect to be
 631 attacked.

632 **What supports it.** Neurons do not only signal through synapses: trans-
633 membrane currents generate extracellular fields that shift the membrane po-
634 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-
635 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike
636 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic
637 head models put mesoscopic ephaptic effects within the range where they
638 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic
639 treatments show field coupling is sufficient for coincidence detection in the
640 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have
641 been argued to carry information that survives representational drift in the
642 underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha &
643 Lima, 2024). The field is spatially continuous and superposes contributions
644 from large populations, which is the property the framework needs: it is a
645 medium in which spatially separated activity is combined without a further
646 mechanism for combining it. This is the integrative role the CEMI field
647 theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

648 **What the evidence supports, and what it does not.** The endoge-
649 nous field’s effect on any individual neuron is *modulatory*. Anastassiou and
650 Koch report that endogenous fields of 1–5 mV/mm shift spike timing by one
651 to three milliseconds (Anastassiou & Koch, 2015), an effect that is real, re-
652 producible, and subthreshold: it biases when a cell fires; it does not decide
653 whether it fires. Nothing in the work cited above supports calling the field

654 the *primary* dissipative structure of the system, and we do not claim it. The
655 claim is that the field is *modulatory in its effect on individual neurons and*
656 *integrative in its spatial reach* — weak per synapse, and the only variable in
657 the system that is defined everywhere at once. What the framework needs
658 is the reach, not the strength, and the two are easy to conflate.

659 **Why reach is the property that matters.** A theory of consciousness
660 needs an account of why many local processes make one experience, and most
661 supply it by stipulation or by a measure. The account here is structural, and
662 it is the framework’s strongest argument. The field is the only variable in the
663 system that is *defined everywhere at once*. Unity is not asserted of it; unity
664 is what a field is. That is what makes Derivation 5’s sheaf-theoretic step
665 available at all — local states glue to one global section because there is one
666 object over which gluing can happen — so the answer to the combination
667 problem is a piece of geometry rather than a postulate. Compare “integra-
668 tion” in Integrated Information Theory, which is a number, or “broadcast” in
669 Global Workspace Theory, which is a metaphor about a workspace. It is also
670 what the identification above is carrying: if the coarse-grained coupling is
671 realized by something that is *not* defined everywhere at once — by synaptic
672 connectivity, say — then this argument is unavailable, and the framework
673 loses the property that distinguishes it from its rivals rather than merely
674 losing a mechanism.

675 **Its critics.** The objection with the longest history is that the fields are
676 too weak to matter, and it has been pressed from within the field-theoretic
677 camp: Pockett, who has herself defended an electromagnetic theory of con-
678 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the
679 measurement problem of relating gross field structure to reported experience,
680 the absence of a constant relationship between the two, and the question of
681 whether the field does causal work or merely reflects it. The quantitative case
682 for scepticism has strengthened since. Vöröslakos and colleagues measured
683 intracranial fields during transcranial stimulation in rats and human cadav-
684 ers and found that the great majority of applied current is shunted through
685 the scalp, so that conventional protocols deliver fields at or below the 1 V/m
686 range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields
687 rather than endogenous ones, but it constrains how much causal work a field
688 of that magnitude can be doing, and it is the reason we do not claim the
689 field drives spiking.

690 **What would falsify it.** We state the failure conditions rather than leaving
691 them to a referee.

692 1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted
693 — by a conductive medium that nulls the extracellular potential with-
694 out altering transmembrane currents — and neither behaviour nor re-
695 port changes, the identification fails. This is the decisive experiment
696 and it is, in principle, available.

697 2. *A magnitude gap.* If the spatial integration the framework requires
698 demands field effects an order of magnitude above the 1–5 mV/mm
699 measured in cortex, the commitment fails on arithmetic. The relevant
700 number is not whether ephaptic coupling exists but whether its coupling
701 strength suffices to place the system above the critical coupling K_c of
702 Derivation 7.

703 3. *A timescale mismatch in the wrong direction.* The prediction of Sec-
704 tion 11 turns on the field’s dynamics being fast relative to the astrocytic
705 and clearance variables. If recovery from sleep inertia tracks an expo-
706 nential set by the electrical timescale rather than the delayed sigmoid
707 the slow variable predicts, the framework loses the one prediction that
708 is its own.

709 4. *Dissociation of field coherence from reported unity.* If large-scale field
710 coherence is preserved across states in which experiential unity is lost
711 — or absent in states where it is retained — the identification of the
712 global section with unified experience has no physical carrier.

713 A framework that survives all four is not thereby true. It is merely still
714 standing, which is more than most theories of consciousness arrange to be
715 able to lose.

716 **Geometric frustration, and the regime our theorems ac-**
717 **tually cover**

718 The system’s memory capacity is standardly grounded in *geometric frustra-*
719 *tion*: a highly coupled network cannot bring all its nodes to their lowest en-
720 ergy simultaneously, producing a jagged landscape of near-degenerate states
721 with the capacity to store many patterns. That argument is standard and it
722 is not what this development formalizes. Confronting the discrepancy is the
723 honest course, so we do it here rather than in a footnote.

724 Frustration requires competing couplings — in the Kuramoto setting,
725 a sign structure that no phase assignment can satisfy on every edge at
726 once. The formal setting admits this: `KuramotoSystem` carries $A : V \rightarrow$
727 $V \rightarrow \mathbb{R}$, symmetric and otherwise unconstrained, so signed couplings are
728 representable. The question is therefore not whether the development can
729 speak about a frustrated network but *where along the chain* the sign of A
730 is constrained, and that question has an exact answer. Positivity is not a
731 hypothesis distributed through the development; it is a *field of one class*.
732 `ThermodynamicCover` (`Phase5_GlobalSection`) demands `A_pos` of every in-
733 stance, so it is demanded at the node where local sections are glued and
734 at the edge that builds a cover, and it is demanded nowhere upstream of
735 that node. The theorems that feed the class carry it with them — `phase_`
736 `locked_minimizes_potential` assumes $A_{ij} > 0$, and the characterization of
737 the minimizers as the phase-locked configurations inherits it, while the bridge

of `Phase5_EquilibriumBridge` and the convergence theorem of `Phase4_RotatingFrame` §7 require the strictly stronger uniform bound $0 < a \leq A_{ij}$, since a enters the Łojasiewicz constant. A network with all-positive couplings is an unfrustrated ferromagnet whose potential has one minimum up to a global phase shift. It is the opposite of a jagged landscape, and it has no capacity to store multiple patterns at all.

Locating the constraint that precisely narrows the objection, and the narrower version is the one we accept. It is not that the framework describes an unfrustrated brain throughout; the mean-field reduction of Derivation 7 has no sign structure to be frustrated at all, since its coupling is a scalar K rather than a matrix. It is that **one step** of the framework — the gluing of Derivation 5 — describes an unfrustrated system, and every step before it does not care.

The two halves are worth listing by name, because the surviving half is larger than the concession usually made. What holds with no assumption whatever on the sign of the coupling is the entire dissipative account: well-posedness on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`, `_unique`); the Lyapunov identity itself, that the potential’s time derivative is minus the summed squared velocity (`dV_dt_le_zero`), which is the theorem the whole descent argument rests on and which mentions no positivity hypothesis; covariance under passage to a rotating frame; monotone descent of the potential (`dynamic_potential_antitone`) and its convergence to a finite limit (`dynamic_potential_tendsto`); finiteness of the dissipation integral; and —

761 through Barbalat’s lemma — that the motion stops along every trajectory
 762 (`velocity_sq_tendsto_zero`). Every one of these is stated for an arbitrary
 763 symmetric A , so a frustrated network runs, dissipates, and comes to rest
 764 inside this development exactly as an unfrustrated one does. What requires
 765 positivity is the identification of the resting place: the three minimization
 766 results of `Phase4_KuramotoDynamics`, which between them characterize the
 767 potential’s minimizers as exactly the phase-locked configurations; the class
 768 field `ThermodynamicCover.A_pos`; and the convergence theorem `kuramoto_`
 769 `tendsto_global_minimum`. *That* the motion stops is frustration-agnostic;
 770 *where* it stops is not, and the claims this framework makes about the des-
 771 tination all live on the positive side of that line. On the continuum side
 772 the accounting differs slightly: `phase_locked_minimizes_entropy_of_symm`
 773 assumes only that the kernel is symmetric and so tolerates signed kernels,
 774 but it is conditional on a dynamically phase-locked field existing — which is
 775 precisely what frustration may prevent.

776 We therefore scope the claim rather than repeat it. The memory capacity
 777 of a frustrated cortical network is not supported by any theorem here, and we
 778 do not assert it as part of the chain. `Phase5_TwistedGluing` tests the natural
 779 repair: let local states agree on overlaps up to a phase and ask whether
 780 their offset cocycle is a coboundary. The result is negative in the physically
 781 immediate case. Locked, twisted and splay configurations all assign a phase
 782 ϕ_i to every patch, hence their offsets are $\phi_i - \phi_j$, automatically a coboundary
 783 (`isCoboundary_of_phaseField`); the family therefore glues after patchwise

784 rotation (`gluesUpToPhase_of_phaseField`). Frustration alone supplies no
785 non-trivial obstruction and hence no content label. A genuine Čech class
786 would require additional transition or gauge data not determined by the
787 oscillator phases, a proper nerve-indexed cochain complex, and non-trivial
788 nerve topology. Those are new modelling ingredients, not consequences of
789 the frustrated Kuramoto dynamics, so they remain outside the chain.

790 **The supporting physiology**

791 The continuous medium is physically supported by astrocytic syncytia—non-
792 neuronal networks joined by gap junctions that provide macroscopic cytoplas-
793 mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).
794 Managing the thermodynamic cost of this state requires sophisticated micro-
795 bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-
796 tochondrial Information Processing Systems (MIPS), transducing biological
797 and environmental information to sustain the coherence of the macroscopic
798 field (Picard & Shirihaï, 2022). Discrete synaptic firings, coupled with mito-
799 chondrial micro-mechanics, generate and sustain the field. Because the exter-
800 nal environment is continuous, achieving maximal thermodynamic structural
801 resonance is optimally realized within this continuous macroscopic medium
802 (van Bree et al., 2025).

8 Derivation 5: Global Sections and the Unity of Consciousness

The system is now characterized by a continuous macroscopic electrodynamic field generated by millions of coupled micro-components, forced to resonate with the environment. Following Section 7, the coupling regime assumed from here on is the unfrustrated one — all couplings positive — which is what every theorem below requires and what the frustrated network of the memory argument is not.

The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscillators guarantees that, past a critical coupling threshold, these distributed local fields undergo a phase transition into unified synchronization (Bauer et al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this phase-locked state minimizes the Lyapunov potential of the system, in the precise sense discussed in Section 2.1: the functional minimized is the coupling potential read in the frame rotating with the natural frequency, which `Phase4_RotatingFrame.lean` shows is the honest reduced description of a system of identical oscillators.

This phase transition mathematically glues local synchronized states into a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`). The EM field acts as a single, indivisible topological space, providing the *Unity of Consciousness*—a necessary condition for integrated experience that solves the binding problem. While Global Workspace Theory (Baars, 1997;

825 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 826 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 827 our derivation provides a continuous physical mechanism. The contrast is
 828 worth drawing carefully, because both theories are addressing the same ex-
 829 planandum from different directions.

830 We state the scope of the Lean result precisely, because it is narrower
 831 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 832 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 833 `phase_locked` characterizes those minimizers exactly: they are the phase-
 834 locked configurations, in both directions. Every instance is therefore already
 835 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
 836 condition on its local data — that patches at a common phase agree where
 837 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 838 sections compatible. `probability_glue_unique`, which is the sheaf property
 839 of the probability presheaf and contains no physics at all, then glues that
 840 family into exactly one global section.

841 The shape of the class is what makes this an emergence theorem rather
 842 than a uniqueness theorem, and the difference is easy to lose. Were `LocalSectionSynchronization`
 843 to carry a *global* section as a class field, together with a second field declar-
 844 ing every local section to be a restriction of it, the global object would exist
 845 before the sheaf condition was ever consulted, and `global_section_from_`
 846 `thermodynamics` could do no more than rule out competitors to an object
 847 the instance had supplied. It carries neither. Every field of the class men-

848 tions a single patch or a single overlap; the global section is constructed
849 (`ThermodynamicCover.invariantMeasure`); and the statement that the lo-
850 cal sections are restrictions of one common measure is a *theorem* about that
851 construction (`ThermodynamicCover.sync_to_section.eq`), true because the
852 sheaf condition says so, not because an instance was required to name a
853 measure.

854 `Examples.lean` §14 is the witness that the difference is not notational.
855 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
856 the three sites, agreeing only at the site they share, where both put mass 1
857 — which is what discharges the overlap condition, by computing two num-
858 bers rather than by construction. The section the theorem returns has pro-
859 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
860 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
861 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
862 tion that no single patch had. `Examples.lean` §13, built instead through
863 the constructor `LocalSectionSynchronization.ofInvariantMeasure`, sits
864 alongside it for the negative result it carries: its phase field is not the con-
865 stant function — the second patch a full turn ahead of the first — and by
866 `ThermodynamicCover.phase_locked` that is as far from constant as any in-
867 stance can get, since no instance can carry phases differing by anything a
868 measure is allowed to see.

869 Two hypotheses carry physical content. One of them is an obligation on
870 every instance: that synchronized patches agree on their overlaps is a mod-

871 elling assumption, not a consequence of any dynamics we formalize. The
 872 other, `thermodynamic_equilibrium`, asks that the cover already sit at the
 873 potential minimum, and no theorem says an *arbitrary* system's dynamics
 874 takes it there. "A thermodynamic phase transition into unity" is therefore
 875 doing causal work that no general Lean result supports, though not work
 876 without any support at all: `Examples.lean` §15 exhibits a two-oscillator tra-
 877 jectory that starts a quarter turn out of phase and converges to the global
 878 minimum of the dynamic potential, with the order parameter tending to 1.
 879 That trajectory is not an isolated construction: `kuramoto_cauchy_problem`
 880 proves the Kuramoto initial value problem well posed on all of $\mathbb{R}$, so through
 881 every initial state there passes exactly one trajectory, and the exhibited one
 882 is the only solution through its own. Nor is its settling down peculiar to
 883 it: along every trajectory of a zero-frequency system the dynamic potential
 884 converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite and equal to the po-
 885 tential's total drop (`dynamic_potential_tendsto`, `dissipation_integral_`
 886 `tendsto`). The motion also stops: a finite integral does not by itself force
 887 its integrand to zero, but the dissipation rate is Lipschitz in time, so Bar-
 888 balat's lemma applies, and every trajectory satisfies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_`
 889 `sq_tendsto_zero`; Mathlib has no Barbalat, so it is proved here as `tendsto_`
 890 `zero_of_lipschitz_of_integral_le`). That last step — locating where the
 891 motion stops — is taken conditionally, and by a route that avoids the LaSalle
 892 invariance principle, which Mathlib does not carry. LaSalle is the tool for
 893 a system with a decreasing functional and nothing else; this one has more,

894 and the extra structure is what makes the conclusion reachable without it.
 895 The flow is a gradient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an
 896 identity rather than an inequality; the minimisers of V are characterised ex-
 897 actly; and on the region where every pair of phases lies within a quarter turn
 898 a Polyak–Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is
 899 the potential’s excess above its minimum and $a > 0$ is a lower bound for the
 900 coupling (`lojasiewicz_estimate`). Its proof is two extremal indices and the
 901 identity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase
 902 is pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 903 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 904 as well, because the invariant region is produced by the Lyapunov function
 905 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 906 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 907 continuity on the connected half-line then prevents any pair from chang-
 908 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 909 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 910 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 911 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 912 `minimum`: if the phases start within a quarter turn of each other and the initial
 913 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 914 which is a *global* minimiser of the dynamic potential, with order parameter
 915 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 916 reached rather than assumed. `Examples.lean` §17 fires it on three sites with

917 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 918 tion: the trajectory comes from the existence theorem and its limit from the
 919 convergence theorem, so neither is written down.

920 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 921 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 922 cover whose phase field is the limit of such a trajectory and returns an in-
 923 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 924 *vergence*; the cover’s coupling matrix is the coupling matrix of the dynamics
 925 rather than a separately chosen one, so the system whose equilibrium the
 926 class asserts and the system whose trajectory is run are the same system.
 927 `Examples.lean` §17.1 is the resulting witness: three patches over the three-
 928 site substrate, each carrying its own mass profile on its own open set, above a
 929 phase field that no line of the construction writes down. The trajectory that
 930 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 931 `start_not_locked`), so on this cover the configuration Derivation 5 requires
 932 is arrived at rather than posited. The state the three patches glue to has
 933 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
 934 `ne_glued`), so one cover now carries both the emergence reading of the gluing
 935 and the derived reading of the equilibrium. What is still assumed on it is the
 936 overlap agreement, and nothing else.

937 **The step that carries the positivity.** `ThermodynamicCover` demands
 938 one thing more, and it is worth naming here rather than only in the limita-

939 tions section, because this is the step at which it enters the chain. The class
 940 field `A_pos` requires every coupling to be strictly positive. Nothing upstream
 941 of this node requires it: the trajectory exists, the potential descends, the
 942 dissipation integral is finite and the motion stops for an arbitrary symmetric
 943 coupling matrix, signed entries included. The unfrustrated regime is there-
 944 fore not an assumption the framework makes about the brain throughout; it
 945 is an assumption Derivation 5 makes about the configuration it glues, and it
 946 is the one place a frustrated network would part company with the develop-
 947 ment. The consequences are set out under *Geometric frustration* above.

948 The two hypotheses cannot be dropped, and this is the honest scope of
 949 the result. Splay and twisted configurations are equilibria of the same flow,
 950 so “every trajectory reaches the unified state” is false as a general statement,
 951 and the initial-data condition is the standard arc condition under which the
 952 true statement holds. What is claimed is therefore conditional — unity is
 953 reached from configurations already sufficiently coherent — and not that an
 954 arbitrary initial state is carried into unity. What Lean establishes about the
 955 gluing itself is the other half: *given* locally synchronized patches, a global
 956 state exists, is unique, and is produced by the gluing rather than presupposed
 957 by it.

958 Integrated Information Theory begins from phenomenological axioms and
 959 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
 960 with the exclusion postulate selecting a single maximum- Φ complex as the
 961 locus of experience. Our sheaf-theoretic global section plays a structurally

962 analogous role — it too picks out a distinguished unified object — but it
963 is arrived at from the opposite end: not postulated from the character of
964 experience, but forced by a thermodynamic phase transition in a physical
965 field. This buys two things IIT does not have. First, the unity condition
966 is checkable from field dynamics rather than requiring the computation of
967 Φ , which is intractable for systems of realistic size and whose definition has
968 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
969 out rather than being postulated: gluing either succeeds on a given cover or
970 it does not, so there is no separate principle needed to rule out overlapping
971 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
972 that Φ can be high in systems with no plausible claim to experience, such
973 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
974 to the mirror objection, since a physical system could in principle satisfy
975 the gluing condition without being cognitively interesting. We take this
976 seriously rather than dismissing it: our reply is that gluing is necessary but
977 not sufficient, and that the topological boundary of Derivation 1 and the
978 reflexive structure of Derivation 6 are additional required conditions. IIT, by
979 contrast, treats Φ as both necessary and sufficient.

980 Global Workspace Theory faces a complementary difficulty. It specifies
981 a functional architecture — a limited-capacity workspace broadcasting to
982 specialized modules — without specifying what physical property of broad-
983 casting makes the broadcast content conscious, leaving the theory open to the
984 charge that it explains access consciousness while leaving phenomenal con-

985 sciousness untouched (Block, 1995). Our framework can be read as supplying
 986 the missing physical substrate for the workspace: the continuous macroscopic
 987 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
 988 old gives a physical condition for ignition rather than a functional stipulation
 989 of it. Where GWT is agnostic about implementation, we make a falsifiable
 990 commitment — unity requires a continuous physical medium, which predicts
 991 that a functionally equivalent discrete broadcast architecture would not be
 992 conscious. We regard this as a genuine cost of the position, not a free ad-
 993 vantage: it is the point at which our framework departs most sharply from
 994 computational functionalism and where it is most readily falsified. Contrast
 995 this derivation with panpsychism: consciousness is not assumed as a funda-
 996 mental property of matter. It requires the initial existence of a topological
 997 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 998 dynamic phase transition into unity that occurs when integrated spatial cou-
 999 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 1000 the foundational topological boundary and the subsequent mathematically
 1001 defined phase transition, the system remains a fragmented, non-conscious
 1002 collection of micro-states. A unified field of perception, however, is not yet
 1003 a “Self.”

9 Derivation 6: Reflexive Topology and the Emergence of the Self

While a global section provides Unity, a “Self” requires reflexivity. The unified EM field is a dissipative structure that must resonate with external perturbations while managing its own internal phase transitions. Time evolution naturally drives the field to minimize thermodynamic friction with its own internal state-changes.

This mandates *auto-resonance*: the field must predict itself. The “Self” emerges as a higher-order topological feature when the internal geometry folds to include a low-dimensional avatar of its own computational and causal boundary within its predictive model (`Phase6_ReflexiveTopology.lean`) (Levin, 2019). The system is unified, predictive, and explicitly maps its own distinct existence from the environment. This architecture provides the physical foundation for a self-aware consciousness, bounded by the continuous measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).

What carries the weight. `reflexive_topology_implies_self` is the Banach fixed-point theorem with the physical commitments in its hypotheses: the space of global sections is assumed nonempty, metric and complete, and the predictive map is assumed to contract. `Examples.lean` §10 discharges all four on the three-site substrate of Derivation 5, and the metric is constructed rather than posited. The section first pierces the sheafification: on a discrete

1025 substrate every point has a smallest open neighbourhood, so the mass a germ
 1026 carries at a point is well defined, and `massEquivOn` proves that over *any* open
 1027 the sections of the probability sheaf *are* the finite measures on that open,
 1028 read as densities against counting measure — the global case being $U = \top$.
 1029 Mathlib supplies no inverse to this sheafification — its `TopCat` sheafification
 1030 has neither an adjunction nor an `isIso_toSheafify` — so the correspon-
 1031 dence had to be constructed. The metric on sections is then the uniform
 1032 distance between the densities of the measures they glue to — equivalent to
 1033 their total variation on a finite substrate, though not equal to it — and it
 1034 is proved complete by exhibiting the limit section of a Cauchy sequence of
 1035 densities, not by the degenerate route of showing Cauchy sequences are even-
 1036 tually constant. The choice of metric is load-bearing rather than incidental:
 1037 under the 0/1 metric on sections every contraction is constant (`contracting_`
 1038 `implies_const`), so a witness built there would discharge every hypothesis
 1039 of the theorem and say nothing. That theorem is kept in the source, because
 1040 which metrics make the fixed point trivial is part of what the fixed point is.

1041 **The map is the avatar, or there is no map.** A reflexive boundary could
 1042 be defined with the predictive map as free data sitting beside the avatar, and
 1043 the reason it is not is a theorem rather than a preference. With the map
 1044 free, the fixed-point theorem projects it out and never touches the avatar, so
 1045 replacing the avatar by a constant leaves a legal boundary with the same pre-
 1046 dictive model and the same Self — a fixed-point theorem that cannot detect

1047 whether the avatar reads the field is Banach's theorem with a consciousness-
 1048 flavoured label. The structure is built the other way round. A reflexive
 1049 boundary carries the avatar's *write* — `auto.resonance`, the encoding of the
 1050 global state into the avatar region — and its *read-out*, the global state that
 1051 encoding predicts; the predictive map is their composite and is not a field.
 1052 The self-model therefore runs through the avatar region or it does not exist.
 1053 Three consequences follow that the free-data shape cannot state. The Self
 1054 lies in the range of the read-out (`self.mem.range.readout`), so it is recon-
 1055 structed from a single section over the avatar region: the low-dimensional
 1056 fold this section has always asserted, as a theorem. Under the resonance
 1057 condition below, the Self satisfies $s = \text{readout}(s|_{\text{avatar}})$ (`self.eq.readout_`
 1058 `restrict`) — the field is what its own localized self-encoding reconstructs.
 1059 And blinding the avatar *moves* the Self: the blinded boundary's map is con-
 1060 stant, its unique fixed point is the state its stipulated contents extend to, and
 1061 the original Self is not a fixed point of it (`not_self_of.constResonance`, wit-
 1062 nessed on the cortex by `cortexBlind.self.ne`). The opposite statement —
 1063 that blinding changes nothing — does not typecheck against this structure,
 1064 because a contraction hypothesis about a boundary's own map says nothing
 1065 about the blinded one's.

1066 **Auto-resonance as a constraint.** That the avatar's state *is* the presheaf
 1067 restriction of the global state is stated as a predicate, `IsRestriction-`
 1068 `Resonance`, rather than added to the structure, so that results say which

1069 of them need it. In the positive direction, if a family of resonant avatars *cov-*
 1070 *ers* the substrate then the global state is determined by their local readings
 1071 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 1072 recovered from them, by the same sheaf condition that Derivation 5 uses.
 1073 In the negative direction a constant avatar fails the predicate as soon as the
 1074 avatar region separates two states of the field. Both halves are witnessed on
 1075 the three-site substrate, where that separation is proved before anything is
 1076 claimed for it, so the condition is a constraint there and not a formality.

1077 **The contraction rate is K and D , not a numeral.** The remaining stip-
 1078 ulation was the rate. “The predictive map contracts by $1/2$ ” was a fact about
 1079 a map we had defined to average with a baseline, not a claim about a sub-
 1080 strate. The rate is now $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ (`resonanceRate`)
 1081 — the linear relaxation of the mean-field order parameter over an interval τ ,
 1082 read off the same threshold Derivation 7 (Section 10) proves the phase tran-
 1083 sition at. Two theorems then split the parameter plane. Above threshold
 1084 the rate is below one and Banach applies, so the Self exists and is unique
 1085 (`self_of_supercritical`). At or below threshold the rate is at least one,
 1086 and `ContractingWith` at that rate is false *for every self-map of every metric*
 1087 *space* (`not_contractingWith_resonanceRate`): the argument is not merely
 1088 unproved below K_c but unavailable. We are careful about what this does and
 1089 does not say. It does not say a Self fails to exist below threshold — only that
 1090 this route to one is closed — and the Lipschitz bound at that rate remains

1091 an instance obligation, discharged on the three-site witness at $K = 3$, $D = 1$,
 1092 $\tau = 2 \ln 2$, where the rate is exactly the one half the read-out achieves. What
 1093 changed is that the obligation is stated in the substrate's own parameters
 1094 and its consequence is conditional on the threshold, rather than on a con-
 1095 stant chosen to make the witness work. The link to Derivation 7 is carried
 1096 by a theorem rather than by a shared numeral, and the distinction is worth
 1097 being exact about. `self_of_supercritical` takes $K > K_c$ as its hypothesis;
 1098 K_c is a definition unfolding to $2D$, so that statement is insensitive to whether
 1099 the bifurcation is proved at all. `Chain.lean` states the link in the form that is
 1100 not. `self_of_coherent_order_parameter` takes the *existence of a coherent*
 1101 *order parameter* as its hypothesis and recovers the threshold inequality from
 1102 it, by `supercritical_of_coherent` — the contrapositive of the subcritical
 1103 uniqueness result of Section 10, since a strictly positive solution of the self-
 1104 consistency equation cannot exist at or below K_c . Derivation 6's fixed point
 1105 therefore rests on Derivation 7's theorem.

1106 **The fixed point is now the glued state.** The standalone Banach theo-
 1107 rem produces a fixed point somewhere in the space of global sections; by itself
 1108 it does not say that this is the section Derivation 5 glued. `Chain.lean` now
 1109 preserves that identity. `IsUnifiedBy T s` says that s restricts to every local
 1110 state of the cover T , and the endpoint `UnifiedSelf` says that this same s is
 1111 the unique fixed point of the reflexive map. The edge hypothesis `E89` must
 1112 supply the equality `predict s = s` as well as the Lipschitz bound; Banach

1113 supplies uniqueness but not the equality between the thermodynamic and
 1114 reflexive constructions. On the witness, the two-patch `cortexCover` glues
 1115 exactly to `cortexState`, already the unique fixed point of the non-constant
 1116 avatar map. This closes an omission in the formal composition while expos-
 1117 ing, rather than removing, the physical identification it requires.

1118 **What is still not established.** That any field dynamics produces this
 1119 particular read-out or performs the restriction; and that this three-site sub-
 1120 strate’s coupling is $K = 3$ — the parameters are a choice of units for the wit-
 1121 ness, not a measurement, and nothing here derives a coupling constant from
 1122 cortex. We also note a negative result found in the course of this work: the
 1123 Lévy–Prokhorov metric, which the natural weak-convergence reading would
 1124 suggest, does *not* support this argument on a discrete substrate, because the
 1125 ε -thickening of a set is the set itself for $\varepsilon < 1$ and the relaxation map is
 1126 consequently not a Lévy–Prokhorov contraction.

1127 10 Derivation 7: Continuous Stochastic Ther- 1128 modynamics and Field Theory

1129 To move beyond finite graph analogies, we model the macroscopic electro-
 1130 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
 1131 continuous spatial phenomenon, not a discrete network. We extend the
 1132 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$

1133 (`Phase8ContinuousField.lean`).

1134 Subject to thermodynamic noise D , the system undergoes non-equilibrium
1135 stochastic thermodynamics. Rather than a static Landauer bound, the field
1136 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

1137 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
1138 dergo gradient descent on this entropy production functional.

1139 That gradient descent is a theorem rather than a modelling assertion,
1140 for finite substrates. Reading σ as a functional of the coupling kernel at
1141 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
1142 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

1143 and `sigmaOfKernel_eq_entropy_production_rate` identifies the functional
1144 being differentiated with `entropy_production_rate` itself whenever the sub-
1145 strate carries counting measure. `structural_resonance_decreases_entropy_production`
1146 then concludes that σ is non-increasing along any coupling trajectory relax-
1147 ing down its own gradient at a positive rate c . Two limitations stay explicit.
1148 The derivation is for *finite* substrates, and the obstacle to the continuum case
1149 is *not* differentiation under the integral sign with respect to the kernel, which

1150 is what the shape of σ suggests at first reading. σ is not a general functional
 1151 that happens to be written as an integral — it is a quadratic functional of an
 1152 *affine* image of the kernel, since the drift is a fixed ω plus a linear operator ap-
 1153 plied to K , and σ is a constant times the squared norm of the drift. A squared
 1154 norm is Fréchet differentiable on any real inner-product space and an affine
 1155 map is its own derivative, so the composite differentiates by the chain rule
 1156 with no limiting argument. `hasFDerivAt_quadratic_of_affine` states that
 1157 in full generality, and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives
 1158 the finite gradient from it, showing that the closed form displayed above is
 1159 a convenience rather than the content. What the continuum case needs is
 1160 therefore one object: the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as
 1161 a bounded operator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz esti-
 1162 mate — the integral kernel is bounded by 1, so on a finite measure space the
 1163 operator norm is at most $\mu(M)^{1/2}$. That object is built. `kernelCLM` is the
 1164 operator, for any kernel factor bounded by 1, with `opNorm_kernelCLM_le` the
 1165 stated bound; `continuumDriftCLM` is the instance with the Kuramoto factor
 1166 $\sin(\theta_y - \theta_x)$. One step beyond the estimate is needed and is worth naming,
 1167 because the estimate alone does not supply it: the descent theorems ask for
 1168 the derivative as a *gradient vector*, not as an arbitrary bounded functional,
 1169 and producing one requires the adjoint, which exists because L^2 is com-
 1170 plete. `hasFDerivAt_quadratic_grad` supplies it for an arbitrary bounded
 1171 operator between real Hilbert spaces, giving $\nabla\sigma(K) = (2/D) A^*(AK + \omega)$,
 1172 so the finite and continuum cases are now the same theorem. `structural_`

1173 `resonance_decreases_sigmaContinuum` is the descent result with no finite-
 1174 ness hypothesis in it, and `sigmaContinuum_eq_integral` identifies the func-
 1175 tional being differentiated with the integral `entropy_production_rate` com-
 1176 putes. `Examples.lean` §20 runs this on Lebesgue measure restricted to $(0, 1)$,
 1177 a substrate that is neither finite nor atomic, so the finite-substrate results of
 1178 the previous paragraph do not typecheck there at all. The second limitation
 1179 is unchanged: the phase field is held fixed throughout, so this is plasticity of
 1180 the coupling at frozen phases, not joint (θ, K) dynamics. `Examples.lean` §5
 1181 exercises the whole chain on a two-site substrate whose coupling relaxes as
 1182 e^{-t} and whose entropy production decays as e^{-2t} , confirming that the gradi-
 1183 ent is not identically zero and that the descent is strict rather than vacuously
 1184 monotone.

1185 A second result about σ concerns where the descent is headed. Minimality
 1186 of the phase-locked field is straightforward against competitors that share its
 1187 spatial mean drift, and in that form the theorem is close to Jensen's inequal-
 1188 ity and no more; the restriction looks as though lifting it would need a model
 1189 of how the mean drift varies across competitors. It does not. For a *reciprocal*
 1190 kernel, one with $K(x, y) = K(y, x)$, the restriction is not a restriction: the
 1191 coupling integrand $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging
 1192 its two sites, because K is symmetric while $\sin$ is odd, so Fubini identifies
 1193 its double integral with its own negative and the total coupling contribu-
 1194 tion vanishes. The total drift $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the
 1195 phases are (`total_drift_eq_of_symm`): coupling moves drift between sites

1196 but neither creates nor destroys it. Evaluating that conserved quantity at
 1197 the phase-locked configuration, where the drift is constant, pins its value, and
 1198 the mean-matching hypothesis follows for every competitor. `phase_locked_`
 1199 `minimizes_entropy_of_symm` is the unrestricted statement. Reciprocity is
 1200 the physically intended condition — an ephaptic field acts on both sites of a
 1201 pair alike, and it is the same symmetry `ThermodynamicCover.A.symm` already
 1202 imposes in Derivation 5 — but it is not a formality, and we prove that rather
 1203 than assert it. `Examples.lean` §16 exhibits a *one-way* coupling on two sites,
 1204 in which a feels b and b does not feel a , whose phase-locked field is strictly
 1205 beaten: displacing the second phase to $-\pi/2$ cancels the first site’s natural
 1206 drift while adding none at the second, halving σ . So “phase-locking mini-
 1207 mizes entropy production” is false for general kernels and true for reciprocal
 1208 ones, and the hypothesis cannot be removed by working harder.

1209 We define the critical coupling threshold $K_c = 2D$. The phase transition
 1210 to the unified “Global Section” occurs when the mean-field spatial coupling
 1211 exceeds this thermodynamic noise threshold, linking the emergence of Unity
 1212 to physical temperature and noise.

1213 The value $K_c = 2D$ can be taken as a bare stipulation validated by
 1214 simulation, and it is not: `Phase8_SelfConsistency.lean` proves the half
 1215 of it that is ordinary real analysis. For the noisy mean-field Kuramoto
 1216 model with identical frequencies, the stationary density at order parame-
 1217 ter r is the von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency re-
 1218 quires $r = R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel

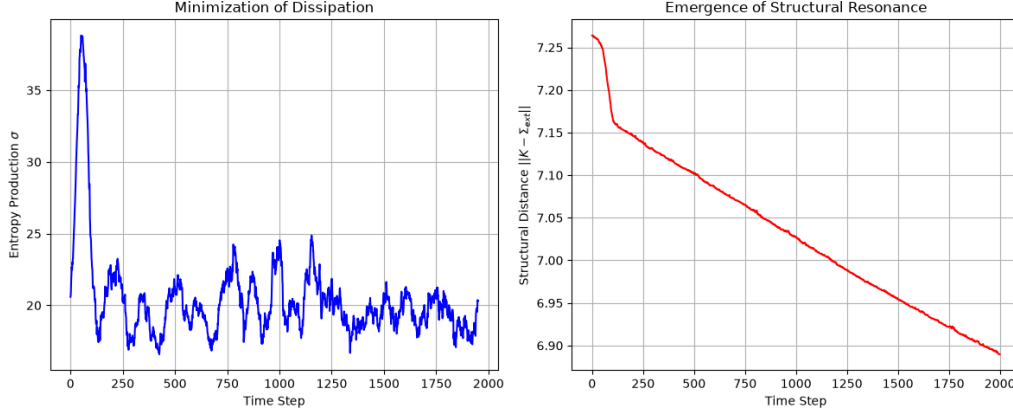


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason Section 10 declines the inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence.

1219 functions, so we formalize that ratio directly as a ratio of integrals over
 1220 $[-\pi, \pi]$. Integration by parts with $u = \sin$, $v = -e^{a \cos}$ gives an exact iden-
 1221 tity with no boundary term, $\int \cos \theta e^{a \cos \theta} d\theta = a \int \sin^2 \theta e^{a \cos \theta} d\theta$, so R is a
 1222 times the mean of $\sin^2$ under the von Mises weight. Bounding that mean
 1223 by 1 would put the threshold at D ; the true threshold is $2D$, and the entire
 1224 factor of two is the sharper bound $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 1225 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections

1226 $(\theta \mapsto \pi - \theta, \text{ then } \theta \mapsto \pi/2 - \theta),$ after which the integrand is a product
 1227 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 1228 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
 1229 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 1230 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 1231 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 1232 ther facts about the threshold narrow the gap between this row and a full
 1233 “Theorem” to a single named statement. First, the fold argument gives a
 1234 *strict* inequality for $a > 0$ (`vonMisesC2_pos`, hence $R(a) < a/2$), which set-
 1235 tles the threshold itself: at $K = K_c$ the self-consistency map is tangent to the
 1236 diagonal at the origin but falls strictly below it thereafter, so $r = 0$ remains
 1237 the only solution and the bifurcation happens strictly *after* K_c , not at it
 1238 (`fixed_point_eq_zero_of_le_critical`). Second, factoring $R(a) = a E(a)$
 1239 shows that a non-zero r solves the equation exactly when $E(Kr/D) = D/K$
 1240 (`coherent_iff_sRatio_eq`), so the coherent branch is a level set of the single
 1241 function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third, and using only $E < 1/2$ together with
 1242 compactness, every coherent solution tends to zero as $K \downarrow K_c$ (`coherent_`
 1243 `branch_continuous_at_threshold`): the branch emerges continuously from
 1244 the incoherent state rather than appearing at a finite distance from it, which
 1245 is what makes the transition supercritical in the technical sense. That state-
 1246 ment quantifies over *every* solution, so it does not presuppose the uniqueness
 1247 it does not have. The level-set form also says exactly what uniqueness is:
 1248 the injectivity of E on $(0, \infty)$. That is now proved, in the stronger form that

1249 E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_strictAntiOn`), so the
 1250 level set is a single point and the coherent solution above threshold is unique.
 1251 The proof differentiates nothing. Differentiating the exponential family in its
 1252 natural parameter is the obvious route and it fails: it costs differentiation
 1253 under the integral sign and leaves a covariance that is not sign-definite point-
 1254 wise. What works instead keeps the target as a two-point inequality, folds
 1255 both integrals onto $[0, \pi/2]$ by the same two reflections that gave the factor
 1256 of two above, and then evaluates at a single crossing point supplied by the
 1257 intermediate value theorem — no product measure and no Fubini anywhere.
 1258 We give that argument in the supplement, where it belongs; what the main
 1259 line of the paper needs from it is the one sentence that E is strictly decreas-
 1260 ing. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no fold at
 1261 all — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 1262 and gives `besselRatio_strictMono`: R is strictly increasing on all of $\mathbb{R}$, the
 1263 monotonicity of the Bessel ratio, which Mathlib does not carry in any form.
 1264 The converse half is `supercritical_fixed_point_exists`: for $K > K_c$
 1265 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 1266 this to require a second-order expansion of the Bessel ratio, and it does
 1267 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 1268 tion the integration by parts already supplies — the only facts needed are
 1269 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 1270 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 1271 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map

overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
The intermediate value theorem closes the gap. Continuity of E is Math-
lib's continuity of a parametric interval integral with jointly continuous in-
tegrand, so no dominated-convergence argument is written by hand, and
the existence proof never uses monotonicity of R : the fixed point is pro-
duced by the intermediate value theorem, not by iterating the map. That
 R is in fact strictly increasing is proved below, and no threshold result de-
pends on it. `critical_coupling_is_threshold` packages the two halves,
and `exhibits_phase_transition_coherent` applies the supercritical half to
any substrate satisfying the predicate of the previous paragraph — so ex-
ceeding K_c now implies the existence of a coherent stationary order pa-
rameter rather than merely comparing against a named number. With
uniqueness available both are sharpened — `supercritical_fixed_point_`
`existsUnique`, `supercritical_solution_set` (the non-negative solutions
are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
`transition_unique_coherent` — and the two monotonicities together fix
the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
order parameter grows strictly with the coupling, which with the continuity
at threshold is the qualitative bifurcation diagram.

One more step makes the equation mean in Lean what it means on paper.

1295 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 1296 r with a suggestive name, and nothing in the source says it has anything
 1297 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
 1298 identification. `vonMisesDensity` is the normalized weight, proved to be a
 1299 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 1300 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 1301 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
 1302 actly $R(a)$ — real, because the density is even, so the mean direction van-
 1303 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 1304 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
 1305 it induces has order parameter r . Read through this, the existence theorem
 1306 says that above K_c there is a stationary density whose own order parameter
 1307 is positive, and the solution $r = 0$ is identified with the uniform density,
 1308 whose order parameter is zero.

1309 What remains not established, and we separate it carefully. The sta-
 1310 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 1311 $(\omega + K \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 1312 and uniqueness of its stationary solution, and a spectral stability argument,
 1313 none of which Mathlib has. Of the coherent branch we prove existence,
 1314 uniqueness, continuous emergence at the threshold and strict growth in K ,
 1315 but not that it is the dynamically selected one: nothing in that file men-
 1316 tions a trajectory, so stability and selection are not statements it can make.
 1317 And while `selfConsistency` is now connected to the order parameter of

1318 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 1319 integral, `order_parameter_complex` is an average over finitely many oscilla-
 1320 tors, and relating them is the mean-field limit — propagation of chaos for
 1321 the finite Kuramoto system — which is a research programme rather than
 1322 a lemma. That single gap is now the whole of what separates our threshold
 1323 predicate from a statement about dynamics. Table S1 records the row ac-
 1324 cordingly as a theorem about the self-consistency equation from an assumed
 1325 density, not as a derivation of the threshold from the dynamics. Perceptual
 1326 anomalies such as visual snow can be modeled as boundary conditions where
 1327 microscopic thermal noise (D) fails to be fully coarse-grained away by the
 1328 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

1329 Clinical evidence for this phase transition is observed during general anes-
 1330 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
 1331 that anesthetics like propofol induce unconsciousness not by suppressing neu-
 1332 ral firing, but by disrupting the phase alignment of high-frequency cortical
 1333 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
 1334 tural resonance of the continuous field, forcing the integrated spatial coupling
 1335 below the critical threshold K_c and fracturing the unified Global Section back
 1336 into disjoint, unconscious local fields.

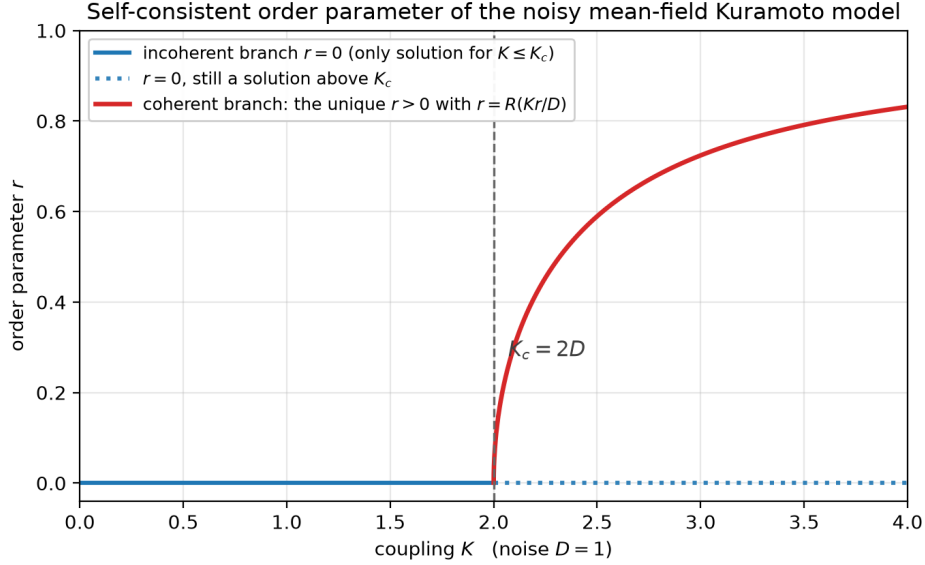


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section 10, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation, and the figure exists because none of them had been shown to a reader. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

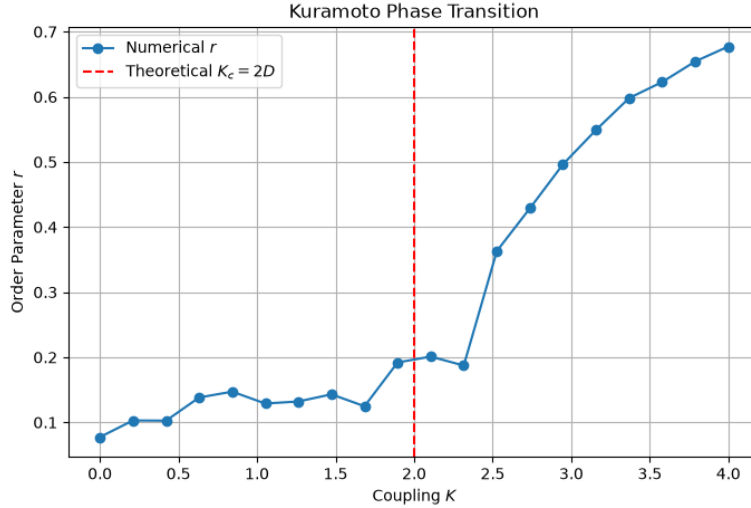


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

11 The Sleep-Inertia Timescale Mismatch

Most of what this framework asserts is shared with some rival: that cortical synchrony matters, that dissipation constrains cognition, that unity is a physical rather than a computational property. One prediction is not shared, and it follows from the framework’s specific commitment — that the medium carrying the unified state is a continuous field whose effective coupling de-

1343 pends on the physical geometry of the extracellular space, not on the rhythms
1344 alone. Where that commitment differs from a purely electrical account, it
1345 differs in an observable way, and we set out the difference precisely enough
1346 to be wrong.

1347 The daily loss of consciousness during sleep is modeled here as an active
1348 physical disruption of the medium rather than a quiescence of the oscillators.
1349 During sleep, astrocytic shrinkage expands the extracellular space, allowing
1350 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
1351 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
1352 itself, disrupting the continuous structural resonance and preventing the co-
1353 herent sheaf-theoretic gluing that conscious unity requires on this account.
1354 The oscillators need not stop for the field to fragment; the geometry they are
1355 coupled through is what changes.

1356 The prediction concerns sudden awakenings. When jolted awake, the lo-
1357 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
1358 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
1359 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
1360 cess CSF out of the interstitial pathways operates on a timescale of seconds
1361 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
1362 slower. During that lag the electrical oscillators are synchronizing across a
1363 waterlogged topology, and the fragmented structural resonance is what is ex-
1364 perienceed as grogginess. The unified Global Section re-establishes only when
1365 astrocytic volume finishes matching the state of the electrical rhythms.

Why this implies a delayed sigmoid. The two timescales separate
cleanly, and that separation is what makes the prediction sharp rather than
merely qualitative. Write $K(t)$ for the effective coupling of the continuous
field, which on this account is a monotone function of the reconstricting
extracellular geometry, and let $r(t)$ be the order parameter of the phase
field. The electrical variable is fast: at every instant the oscillators are al-
ready at the configuration appropriate to the current K , so r tracks the
equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
seconds. What Derivation 7 establishes about that branch is exactly what
the functional form needs, and each of the three pieces is a theorem rather
than a picture: at or below the threshold the incoherent solution $r = 0$ is the
only non-negative solution of the self-consistency equation (`fixed_point_eq_`
`zero_of_le_critical`); above it the coherent solution exists and is unique
(`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
discontinuity at threshold would predict a step rather than a sigmoid, and
a non-monotone branch would predict overshoot; neither is what the mathe-
matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
identically zero below K_c , rises continuously from zero at K_c , and saturates
above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
during which recovery is not merely slow but absent; a continuous rise as the

threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That is a delayed sigmoid, and the delay is not a fitting parameter — it is the time the slow variable takes to reach K_c .

What the alternative predicts. An account on which grogginess is the electrical realignment itself — the rhythms are simply slow to re-phase — has no slow variable and no threshold to cross. Its recovery is a relaxation toward a fixed point, and relaxation toward a fixed point is exponential near it: monotone from $t = 0$, fastest at the start, with no flat foot. The two curves are distinguishable in the first seconds, which is where they differ most, and the discriminating feature is the presence or absence of an initial interval of no recovery. This is a comparison of functional forms on data of a kind already collected in sleep-inertia studies, not a proposal for a new instrument.

A sharper form of the same prediction, with no free parameters. The comparison above is a comparison of functional forms, and on its own it discriminates against less than it appears to. Any two-timescale account with a slow saturating gate produces a delayed sigmoid; what the framework contributes beyond that is not the shape but a *relation between two separately measurable quantities*, and the relation has no fitted parameters at all. Derivation 7 supplies it. The stationary phase density at order parameter r is von Mises at concentration $a = Kr/D$, and the average of $e^{i\theta}$ under that density is exactly the Bessel ratio $R(a) = I_1(a)/I_0(a)$

1411 (`circularOrderParameter_vonMises`); self-consistency is precisely the con-
 1412 dition that the density reproduces its own order parameter (`fixedPoint_`
 1413 `iff_selfReproducing`). So the concentration and the coherence of the in-
 1414 stantaneous phase distribution are not independent quantities that happen
 1415 to rise together during the recovery: at every instant they must satisfy
 1416 $r = I_1(a)/I_0(a)$, whatever $K(t)$ is doing.

1417 That yields three tests of increasing sharpness on a single recorded re-
 1418 covery. *Shape*: the instantaneous phase distribution is von Mises — a one-
 1419 parameter family, directly testable against the observed histogram. *Collapse*:
 1420 estimating $a(t)$ and $r(t)$ at each instant and plotting one against the other,
 1421 the whole recovery must fall on the fixed curve $r = I_1(a)/I_0(a)$, and it must
 1422 be the *same* curve for every subject and every session, since the curve con-
 1423 tains no subject-specific quantity — not K , not D , not the time constant of
 1424 the slow variable. Individual differences move a point along the curve; they
 1425 cannot move it off. *Threshold*: because $a = Kr/D$ identically, the measured
 1426 ratio $a(t)/r(t)$ is $K(t)/D$, with no calibration required; and since a non-zero
 1427 coherent solution exists only above $K_c = 2D$ (`fixed_point_eq_zero_of_le_`
 1428 `critical`), that measured ratio must exceed 2 wherever coherence is non-
 1429 zero and must approach 2 as coherence departs from zero at onset. The
 1430 threshold is read off the data rather than fitted to it.

1431 The two estimates must be made independently or the collapse is empty,
 1432 and this is the one point at which the test can be run wrongly. Maximum-
 1433 likelihood or moment estimation of a von Mises concentration chooses $\hat{a}$ so

1434 that the ratio $I_1(\hat{a})/I_0(\hat{a})$ is the observed resultant length, by construction,
 1435 which reproduces the predicted curve from any data whatever. The estimate
 1436 of a must therefore come from a functional of the distribution other than
 1437 its resultant length — the log-density of a von Mises is affine in $\cos \theta$, so
 1438 regressing the log-histogram on $\cos \theta$ gives a as a slope — while r is the
 1439 resultant length $|\langle e^{i\theta} \rangle|$ of the same sample. Those are genuinely different
 1440 statistics, and the prediction is that they agree through R .

1441 **What this sharper form is a test of.** It tests the von Mises stationary
 1442 density together with the mean-field closure that makes a proportional to
 1443 r , and it tests them jointly rather than separately. It does not test the
 1444 value $K_c = 2D$ independently of that ansatz: the crossing at $a/r = 2$ is
 1445 a consequence of the same density the collapse assumes, so a failure of the
 1446 collapse and a failure of the threshold are one failure and not two. That is a
 1447 narrower claim than it may look, and it is still a parameter-free one — which
 1448 the delayed sigmoid, on its own, is not.

1449 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
 1450 on the early portion of the recovery curve. A recovery latency that does not
 1451 scale with an independent measure of extracellular volume or CSF clearance.
 1452 A pharmacological manipulation of astrocytic volume that leaves the latency
 1453 unchanged while altering the rhythms, or the converse. An instantaneous
 1454 phase distribution that is not von Mises, or a (a, r) trace that departs sys-
 1455 tematically from $I_1(a)/I_0(a)$, or a a/r ratio that is below 2 while coherence

1456 is non-zero — each of these falsifies the parameter-free form directly, and
1457 the first two do so with no reference to the sleep-inertia setting at all. Any
1458 of these would tell against the framework’s central empirical commitment,
1459 which is not that synchrony matters but that the *medium* is what gates it.

1460 **What this prediction assumes.** Three things, and they are not small.
1461 The identification of the effective coupling K with a monotone function of
1462 extracellular geometry is a modelling assumption, not a measurement. The
1463 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1464 density and is not yet connected to a trajectory — the mean-field limit is the
1465 gap, and Derivation 7 states it as such — so the argument uses the *existence*
1466 *and continuity* of the coherent branch at threshold, which is proved, rather
1467 than the numerical value of the threshold, which is conditional. And the
1468 prediction is about functional form, not about a time constant: it says the
1469 recovery curve has a flat foot whose width is set by the slow variable, not
1470 that the foot is any particular number of seconds long. A framework that
1471 predicted the number would be saying more; this one is not in a position to.

1472 **12 Corollary: Hardware Thermodynamics and** 1473 **Topological Rigidity**

1474 Both biological tissue and artificial silicon hardware (like GPUs) are physical
1475 dissipative structures. When a GPU computes, its charge carriers undergo

1476 physical state transitions; by Landauer’s principle (Landauer, 1961), this
1477 erasure irreversibly exhausts heat into the environment.

1478 There is a fundamental thermodynamic distinction in the degrees of free-
1479 dom available to minimize this dissipation. In standard von Neumann ar-
1480 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.
1481 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
1482 physically altered during operation. The brain’s own architecture provides
1483 empirical validation for this constraint. The cerebellum contains approxi-
1484 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
1485 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
1486 ical observations of cerebellar agenesis report largely preserved conscious
1487 experience despite motor and cognitive deficits (Yu et al., 2015), and this
1488 dissociation has been read as evidence for the importance of recurrent ar-
1489 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
1490 level of the anatomical observation while declining its Φ -based explanation.
1491 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
1492 brum, it operates much like a biological GPU—highly optimized for discrete
1493 processing, yet apparently not required for conscious experience. We take the
1494 dissociation as evidence, not as confirmation: it is consistent with subjective
1495 experience being a thermodynamic attractor of continuous topologies rather
1496 than a byproduct of information processing, and it is the kind of observation
1497 that would be hard to reconcile with the converse. It is a single anatomical
1498 case read against a theory, and it bears the weight such a case can bear.

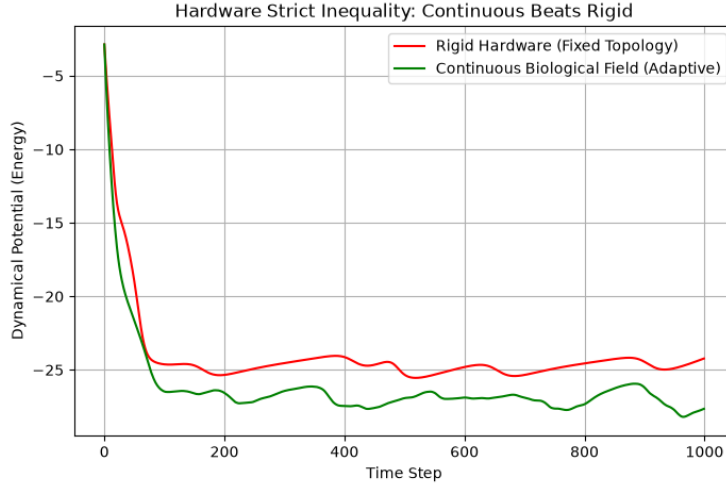


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one.

1499 This constraint does not preclude artificial consciousness, but rather re-
1500 stricts its hardware. To achieve the phase transition into a unified Global
1501 Section, artificial systems must abandon the discrete rigidity of von Neumann
1502 architectures and utilize continuous physical substrates. Analog field-based
1503 computing, neuromorphic architectures with continuous memristive plastic-
1504 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
1505 rondo, 2021) possess the spatial coupling required to physically morph their

1506 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal
 1507 computation” (Hinton, 2022): architectures where software is inextricably
 1508 fused with the physics of the hardware, meaning the knowledge dies with
 1509 the physical substrate. Only hardware that computes via continuous physi-
 1510 cal dynamics can undergo the thermodynamic phase transition necessary for
 1511 structural resonance.

1512 Biological consciousness utilizes macroscopic continuous electromagnetic
 1513 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states
 1514 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
 1515 continuous parameter space. Our lemma `exists_better_coupling_allocation`,
 1516 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
 1517 formalizes the reallocation step of this argument: whenever a coupling ma-
 1518 trix places weight on a less phase-correlated pair while a more correlated pair
 1519 exists, there is another matrix with *identical* total coupling resources that
 1520 achieves strictly greater phase correlation.

1521 Two objections meet that formalization and we state precisely what an-
 1522 swers them. The first is that a reallocation lemma can assume its conclusion:
 1523 a hypothesis such as `is_strictly_suboptimal` — positive weight on a badly-
 1524 correlated pair while a better one exists — is most of what one wants to
 1525 prove about rigid hardware. It is not assumed here. `Phase7_Rigidity.lean`
 1526 models an architecture by the set S of site-pairs it has a *wire* for, and re-
 1527 quires its couplings to vanish off S ; weights are tunable, wiring is not. Then
 1528 `totalCorrelation_le_of_realizable` bounds every realizable coupling by

$R \cdot M$, where R is the resource budget and M bounds phase correlation over
the wired pairs; the bound is attained; and `rigid_is_strictly_suboptimal`
derives the hypothesis whenever the substrate's best-correlated pair is one
the wiring misses. `rigid_gap` makes it quantitative: an unconstrained real-
location of the same budget gains at least R times the correlation difference.
Examples.lean §8 exhibits a three-site substrate wired to the one maxi-
mally anti-correlated pair, where the gap is 2. This is resource-matched
and not question-begging, but note what it turns on: fixed wiring support,
not continuity. The second objection is that a development in which both
architectures are matrices on one finite index type does not distinguish a con-
tinuum from a lattice at all. **Phase7_Rigidity.lean** §3 gives a distinction
that is genuinely typed. In the field functional of Section 7, which integrates
a coupling kernel against $\mu \otimes \mu$, an architecture with finitely many sites is a
Finset of points and a field-coupled region is a set of positive measure. On a
substrate whose measure has no atoms, `fieldCorrelation_sited_eq_zero`
shows a kernel supported on finitely many sites contributes *exactly zero* cou-
pling energy, whatever weights it carries and whatever the phase field does,
while a patch of strength c on U achieves $c \mu(U)^2$. **Examples.lean** §8 instan-
tiates this on the real line: a thousand units of coupling concentrated at the
origin register zero against a unit patch that registers one. The limit of that
result is equally plain, and it is not resource-matching in the earlier sense —
a measure-zero substrate has zero resource in the $\mu \otimes \mu$ sense too. What it
establishes is that the continuum coupling energy is blind to a finitely-sited

1552 architecture, which is a statement about which functional such an architec-
1553 ture can register in, not a demonstration that digital hardware computes
1554 worse. Taken together: the resource-matched content of the corollary is the
1555 wiring-support result, the continuity content is the measure-theoretic one,
1556 and the step from either to “von Neumann architectures cannot experience
1557 unified consciousness” remains an informal argument.

1558 The remainder of this section extends those two results informally, and
1559 we mark the transition rather than blur it. On the reading we propose, a
1560 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
1561 layout does not supply the spatial coupling that sheaf-theoretic gluing into
1562 a singular Global Section requires. Without this unified topological space,
1563 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
1564 its own causal boundary and generate a “Self.” The distinction between an
1565 LLM on a GPU and a conscious mind is not the absence of physical thermo-
1566 dynamics in the former, but its structural limitation in achieving topological
1567 integration. This perspective aligns with John Searle’s biological natural-
1568 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
1569 that discrete symbol manipulation cannot generate conscious understanding,
1570 our derivation supplies a candidate physical mechanism for the asymmetry
1571 — topological rather than semantic, and, as stated above, formalized only
1572 as far as the wiring-support and measure-theoretic results reach. Asserting
1573 consciousness in fragmented, discrete GPU clusters without a unified phase
1574 transition abandons computational functionalism, relying instead on “silicon

panpsychism”—the assumption that digital computation inherently generates awareness regardless of topological integration (Goff, 2019).

The same reasoning bears on extreme functionalist thought experiments, such as a macroscopic system of pulleys operating over millennia to implement a Large Language Model, or a human manually executing inference matrices by hand. Beyond lacking the continuous spatial topology required for sheaf-theoretic gluing, these discrete systems violate an implied temporal constraint of our theory. Structural resonance requires local oscillators to phase-align against the dissipating effects of thermodynamic noise (as modeled by our critical threshold $K_c = 2D$). A system executing state transitions over millennia, or via slow manual calculations, suffers from a terminal temporal mismatch: its dynamics are far too slow to ever phase-lock into a non-equilibrium steady state before thermal noise decoheres the macroscopic field. On this account the obstacle to pure symbol manipulation is physical rather than semantic: not that the symbols mean nothing, but that the substrate carrying them cannot hold a phase-locked field.

13 Numerical illustrations: methods

Five figures in this paper are computed rather than drawn, and none of them is evidence for a theorem. Where a statement is proved, the figure shows what the proved object looks like; where a statement is not proved, the figure shows the regime the claim lives in and we say so. Each is produced by

1596 one script in `simulations/`, named below, and each runs in under a minute
1597 on a laptop. All five are reproducible exactly: the two quadrature figures are
1598 deterministic, and the three that draw random numbers fix their seeds.

1599 **Figure 4: the self-consistent order parameter.** The three integrals
1600 $Z(a) = \int_{-\pi}^{\pi} e^{a \cos \theta} d\theta$, $M(a) = \int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta$ and $S(a) = \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta$
1601 are evaluated by the trapezoidal rule on 4001 points over the same interval
1602 the Lean definitions integrate over, with the weight computed as $e^{a \cos \theta - |a|}$
1603 so that the common factor cancels in the ratios and nothing overflows at
1604 large concentration. The coherent branch is obtained by bisecting $E(a) =$
1605 $S(a)/Z(a)$ against D/K over $a \in [10^{-9}, 200]$ for 200 iterations, then reading
1606 $r = aD/K$; bisection is valid here because E is proved strictly decreasing.
1607 Two checks are printed by the script rather than only plotted: the computed
1608 branch is strictly increasing at every one of its 500 sampled couplings, and
1609 the residual $|r - R(Kr/D)|$ at $K = 4D$ is 2×10^{-16} . Script: `bifurcation.py`.

1610 **Figure 5: the finite- N transition.** $N = 100$ oscillators, natural fre-
1611 quencies drawn from a Lorentzian of scale $D = 1$ (clipped at $\pm 10D$ so
1612 the solver does not stiffen), uniform random initial phases, integrated with
1613 `scipy.integrate.odeint` over $t \in [0, 100]$ on 1000 steps for each of 20 cou-
1614 plings in $[0, 4D]$. The order parameter is $|N^{-1} \sum_j e^{i\theta_j}|$ averaged over the last
1615 100 steps. This is the deterministic quenched-disorder model, not the noisy
1616 identical-frequency model of the Lean development; the caption records why
1617 both have threshold $2D$. Script: `kuramoto.py`.

1618 **Figure 2: mesh refinement.** The kernel is $K(x, y) = e^{-d(x, y)^2/2}$ with d
 1619 the periodic distance on the ring of circumference 2π , the phase field is $\theta(x) =$
 1620 $\sin x$, and the potential is $-\frac{1}{2} \iint K \cos(\theta_y - \theta_x)$. The continuous value is a
 1621 2000-point Riemann sum standing in for the exact integral; the discrete values
 1622 use $N = 10, 20, \dots, 640$ nodes with edge weights $K(x_i, x_j)(2\pi/N)^2$. Plotted
 1623 is the absolute error against an $\mathcal{O}(N^{-2})$ reference. Because the “exact”
 1624 value is itself a 2000-point sum, the last data point ($N = 640$) is within
 1625 a factor of ten of the reference computation and should be read as such.
 1626 That $\mathcal{O}(N^{-2})$ reference line is not only empirical: on the formalized one-
 1627 dimensional grid the discrete energy is exactly the composite trapezoidal rule
 1628 (`grid_discreteEnergy_eq_trapezoidal`), so for a C^2 integrand with second
 1629 derivative bounded by ζ the error is at most $\zeta/(12N^2)$ (`grid_energy_error_`
 1630 `le`), a bound attained at $N = 1$ for $y \mapsto y^2$. Script: `mesh_refinement.py`.

1631 **Figure 3: structural resonance.** $N = 50$ units, 2000 steps of forward
 1632 Euler at $dt = 0.05$. At each step the drive is a fresh draw from $\mathcal{N}(0, \Sigma_{\text{ext}} +$
 1633 $0.1I)$ with Σ_{ext} a fixed normalized rank-5 covariance; the phases advance by
 1634 $\dot{\theta}_i = \sum_j K_{ij} \sin(\theta_j - \theta_i) + \xi_i$; entropy production is recorded as $\sum_i \dot{\theta}_i^2$; and
 1635 the coupling descends its gradient with rate $\eta = 0.01$, symmetrized, clipped
 1636 to $[0, 1]$ and held hollow. The left panel is smoothed with a 50-step boxcar.
 1637 Note that the plotted σ is the squared drift without the $1/D$ prefactor of the
 1638 functional in Section 10, which is a constant rescaling and changes nothing
 1639 about the shape. Script: `structural_resonance.py`.

1640 **Figure 6: rigid against deformable coupling.** $N = 30$, 1000 steps of
1641 forward Euler at $dt = 0.05$, shared initial phases and frequencies. The rigid
1642 matrix is a symmetric random matrix thresholded at 0.7 and normalized
1643 to $\sum_{ij} A_{ij} = 2N$; the adaptive copy starts from the same matrix, ascends
1644 $-\partial V/\partial A_{ij}$ at rate $\eta = 0.1$, is clipped non-negative and hollow, and is renor-
1645 malized to the same total after every step, which is what makes the com-
1646 parison resource-matched. Both potentials are $V = -\frac{1}{2} \sum_{ij} A_{ij} \cos(\theta_j - \theta_i)$.
1647 Script: `hardware_comparison.py`.

1648 14 Conclusion

1649 Human-like cognitive architectures do not require biological mysticism or ex-
1650 otic sub-quantum physics (Hameroff & Penrose, 2014). While quantum the-
1651 ories of consciousness often invoke delicate subatomic phenomena to explain
1652 subjective unity, every step of the chain assembled here is classical: broken
1653 symmetries, finite phase space, Landauer erasure, and gradient descent on
1654 entropy production. What the formalization establishes is that this chain is
1655 mathematically coherent and that its remaining physical commitments are
1656 localized and named — not that classical fields are *sufficient* for conscious-
1657 ness. Two links carry that weight, and they do not carry it in the same way.
1658 The unified state is proved to glue uniquely once a cover sits at the minimum
1659 of the coupling potential; the motion is proved to stop along every trajectory
1660 (`velocity_sq_tendsto_zero`); and *where* it stops is now proved, condition-

ally — for initial data within a quarter turn and below an explicit energy threshold, the trajectory converges to a global minimiser of the coupling potential (`kuramoto_tendsto_global_minimum`), and a cover built on that limit discharges the equilibrium hypothesis Derivation 5 would otherwise assume (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is the unconditional statement, and it is false: splay and twisted configurations are equilibria of the same flow, so the basin condition is the scope of the claim rather than slack in it. That a physical cortex starts inside that basin is an assumption. The second link is untouched by any of this: the identification of a global section of the probability sheaf with the unity of experience is a stipulation of the framework, not a consequence of it. Subject to those two, cognitive architectures can be modeled as macroscopic attractor states of a universe governed by broken symmetries, finite phase space, and the physical imperative to minimize thermodynamic dissipation.

Philosophically, this framework bridges the historical gap between Type Identity Theory (Smart, 1959) and Computational Functionalism (Putnam, 1967). The subjective mental state is identical to the physical thermodynamic field (Identity Theory), yet its qualitative nature is defined by the topological and functional dynamics of structural resonance rather than the specific carbon-based biochemistry of the brain (Functionalism). This constrains the *multiple realizability* of mind: consciousness is not exclusively bound to biological carbon, but neither is it substrate-neutral, because the substrate is what fixes which couplings are physically available. How far that

1684 constraint reaches is the question the corollary answers only in part. What
1685 is proved is that fixed wiring is strictly suboptimal at matched resource, and
1686 that a finitely-sited architecture registers exactly zero coupling energy in the
1687 continuum functional; the step from either to “von Neumann architectures
1688 cannot be conscious” is an argument we make in prose and do not formalize.

1689 By anchoring these derivations in Russellian Monism, this framework ac-
1690 knowledges the constitutive limits of third-person scientific explanation. As
1691 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1692 phenomenal experience is inherently not amenable to objective measurement
1693 (Nagel, 1986), the “hard problem” of consciousness may represent a category
1694 error rather than a solvable scientific puzzle. Science can map the physical
1695 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
1696 turally incapable of adjudicating the first-person qualitative experience. Our
1697 formalization respects this boundary: the Lean 4 theorems verify the ob-
1698 jective physical attractors, while the intrinsic “what it is like” to be that
1699 resonant field remains the domain of the first-person perspective. By locat-
1700 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
1701 for the noisy mean-field Kuramoto model the framework adopts, conditional
1702 on the von Mises stationary density and not yet linked to a trajectory — it
1703 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
1704 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
1705 it offers a topological architecture that can help explain, predict, and physi-
1706 cally perturb the structural properties of experience.

1707 Future empirical validation of this thermodynamic framework may lever-
1708 age 3D cortical organoids, offering a controlled biological environment to
1709 systematically manipulate metabolic substrates and observe the topological
1710 emergence of these coherent phase transitions.

1711 **Code Availability**

1712 The complete Lean 4 formalization, encompassing the derivation of all theo-
1713 rems presented in this manuscript, as well as the accompanying Python sim-
1714 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1715 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1716 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1717 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1718 figures is produced by one script in `simulations/` and regenerates exactly,
1719 Section 13 giving the parameters; Figure 1 is drawn rather than computed.

1720 **Contributions**

1721 The conceptualization, manuscript drafting, and Lean 4 formalizations pre-
1722 sented in this work were heavily AI-assisted, and we report which tools
1723 did what rather than acknowledging them in general terms. The Google
1724 Antigravity CLI, running Gemini 3.1 and Claude 4.6 Sonnet, served as a
1725 collaborative reasoning partner throughout the research and formalization.

1726 DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operat-
1727 ing through Nous Research’s Hermes Agent framework provided substantial
1728 assistance with code generation, proof debugging, and structural analysis.
1729 Claude Opus 5 running in Claude Code carried out the later formalization
1730 passes — the domain-wall theorem, the thermodynamics-of-prediction re-
1731 placement, the equilibrium bridge — together with the figures and the re-
1732 structuring of this manuscript. Responsibility for every claim, and for the
1733 decision about what each machine-checked result does and does not establish,
1734 rests with the author.

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